



DRV8701 有刷直流电机全桥栅极驱动器

1 特性

- 单路 H 桥栅极驱动器
 - 驱动 4 个外部 N 通道金属氧化物半导体场效应晶体管 (MOSFET)
 - 支持 100% 脉宽调制 (PWM) 占空比
- 5.9V 至 45V 工作电源电压范围
- 两个控制接口选项
 - PH/EN (DRV8701E)
 - PWM (DRV8701P)
- 可调节栅极驱动 (5 级)
 - 6mA 至 150mA 拉电流
 - 12.5mA 至 300mA 灌点流
- 支持 1.8V、3.3V 和 5V 逻辑输入
- 分流放大器 (20V/V)
- 集成 PWM 电流调节功能
 - 限制电机浪涌电流
- 低功耗睡眠模式 (9μA)
- 两个低压降 (LDO) 稳压器, 用于为外部元件供电
 - AVDD: 4.8V, 最高支持 30mA 输出负载
 - DVDD: 3.3V, 最高支持 30mA 输出负载
- 小型封装尺寸
 - 24 引脚超薄型四方扁平无引线 (VQFN) (PowerPAD™)
 - 4.0mm × 4.0mm × 0.9mm
- 保护特性:
 - VM 欠压闭锁 (UVLO)
 - 电荷泵电压 (CPUV)
 - 过流保护 (OCP)
 - 前置驱动器故障 (PDF)
 - 热关断 (TSD)
 - 故障状态输出 (nFAULT)

2 应用

- 工业有刷直流电机
- 机器人
- 家庭自动化
- 工业泵和阀
- 电动工具
- 手持式真空吸尘器

3 说明

DRV8701 是一款采用 4 个外部 N 通道 MOSFET 的单路 H 桥栅极驱动器, 主要用于驱动 12V 至 24V 双向有刷直流电机。

该器件可通过 PH/EN (DRV8701E) 或 PWM (DRV8701P) 接口轻松连接控制器电路。内置的感测放大器能够实现可调节的电流控制。这款栅极驱动器内置有相应的电路, 以便能够采用固定关断时间的 PWM 电流斩波来调节绕组电流。

DRV8701 采用 9.5V V_{GS} 栅极驱动电压来驱动高侧和低侧 FET。所有外部 FET 的栅极驱动电流均可通过 IDRIVE 引脚上的单个外部电阻进行配置。

低功耗睡眠模式可将内部电路关断, 从而实现极低的静态电流消耗。这种睡眠模式可通过将 nSLEEP 引脚置为低电平来设定。

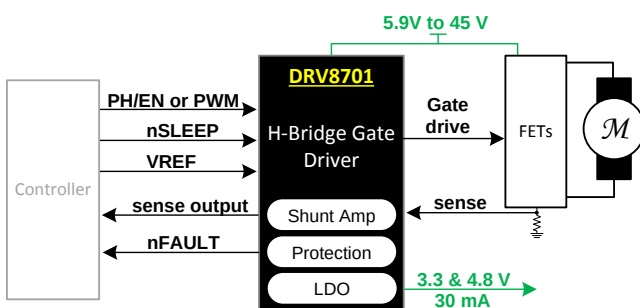
该器件内置以下保护功能: 欠压锁定、电荷泵故障、过流关断、短路保护、前置驱动器故障以及过热保护。故障情况通过 nFAULT 引脚指示。

器件信息⁽¹⁾

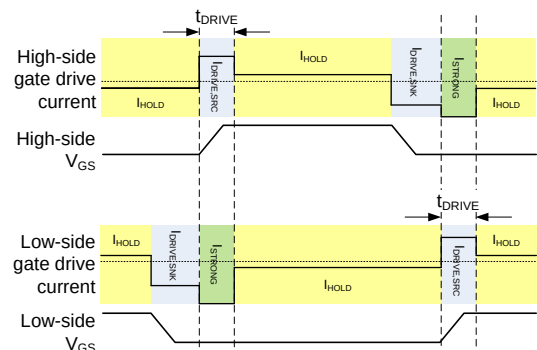
器件型号	封装	封装尺寸 (标称值)
DRV8701	VQFN (24)	4.00mm × 4.00mm × 0.90mm

(1) 要了解所有可用封装, 请见数据表末尾的可订购产品附录。

简化的系统框图



栅极驱动电流



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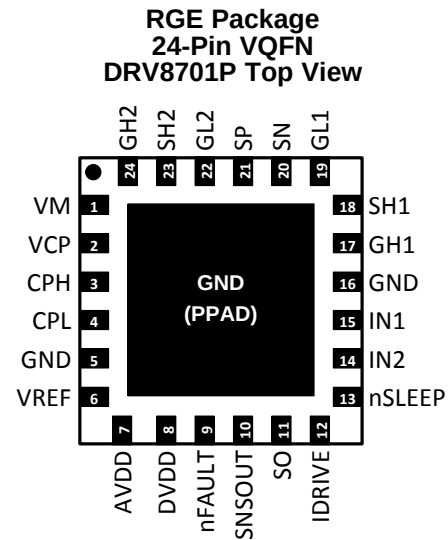
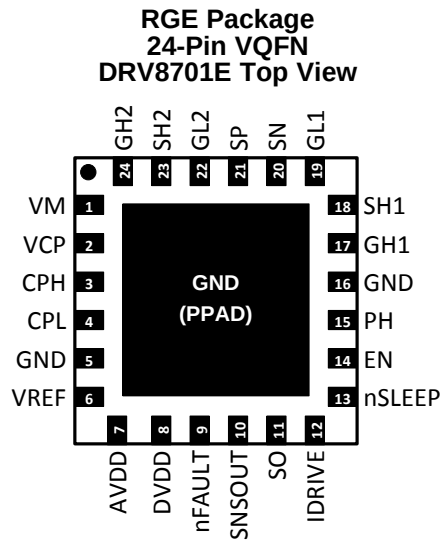
4 修订历史记录

Changes from Original (March 2015) to Revision A

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• 已将器件状态更新为量产数据	1
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5 Pin Configuration and Functions



DRV8701E (PH/EN)

PIN		TYPE	DESCRIPTION	
NAME	NO.			
EN	14	Input	Bridge enable input	Logic low places the bridge in brake mode; see Table 1
PH	15	Input	Bridge phase input	Controls the direction of the H-bridge; see Table 1

DRV8701P (PWM)

PIN		TYPE	DESCRIPTION	
NAME	NO.			
IN1	15	Input	Bridge PWM input	Logic controls the state of H-bridge; see Table 2
IN2	14	Input		

Common Pins

PIN		TYPE	DESCRIPTION	
NAME	NO.			
VM	1	Power	Power supply	Connect to motor supply voltage; bypass to GND with a 0.1-μF ceramic plus a 10-μF minimum capacitor rated for VM; additional capacitance may be required based on drive current
GND	5	Power	Device ground	Must be connected to ground
	16			
	PPAD			
VCP	2	Power	Charge pump output	Connect a 16-V, 1-μF ceramic capacitor to VM
CPH	3	Power	Charge pump switching nodes	Connect a 0.1-μF X7R capacitor rated for VM between CPH and CPL
CPL	4			
DVDD	8	Power	Logic regulator	3.3-V logic supply regulator; bypass to GND with a 6.3-V, 1-μF ceramic capacitor
AVDD	7	Power	Analog regulator	4.8-V analog supply regulator; bypass to GND with a 6.3-V, 1-μF ceramic capacitor
nSLEEP	13	Input	Device sleep mode	Pull logic low to put device into a low-power sleep mode with FETs High-Z; internal pulldown
IDRIVE	12	Input	Gate drive current setting pin	Resistor value or voltage forced on this pin sets the gate drive current; see applications section for more details

Common Pins (continued)

PIN		TYPE	DESCRIPTION	
NAME	NO.			
VREF	6	Input	Analog reference input	Controls the current regulation; apply a voltage between 0.3 V and AVDD
nFAULT	9	Open Drain	Fault indication pin	Pulled logic low with fault condition; open-drain output requires an external pullup
SNSOUT	10	Open Drain	Sense comparator output	Pulled logic low when the drive current hits the current chopping threshold; open-drain output requires an external pullup
SO	11	Output	Shunt amplifier output	Voltage on this pin is equal to the SP voltage times A_V plus an offset; place no more than 1 nF of capacitance on this pin
SN	20	Input	Shunt amplifier negative input	Connect to SP through current sense resistor and to GND
SP	21	Input	Shunt amplifier positive input	Connect to low-side FET source and to SN through current sense resistor
GH1	17	Output	High-side gate	Connect to high-side FET gate
GH2	24			
GL1	19	Output	Low-side gate	Connect to low-side FET gate
GL2	22			
SH1	18	Input	Phase node	Connect to high-side FET source and low-side FET drain
SH2	23			

External Passive Components

COMPONENT	PIN 1	PIN 2	RECOMMENDED
C_{VM1}	VM	GND	0.1- μ F ceramic capacitor rated for VM
C_{VM2}	VM	GND	≥ 10 - μ F capacitor rated for VM
C_{VCP}	VCP	VM	16-V, 1- μ F ceramic capacitor
C_{SW}	CPH	CPL	0.1- μ F X7R capacitor rated for VM
C_{DVDD}	DVDD	GND	6.3-V, 1- μ F ceramic capacitor
C_{AVDD}	AVDD	GND	6.3-V, 1- μ F ceramic capacitor
R_{IDRIVE}	IDRIVE	GND	See Typical Applications for resistor sizing
R_{nFAULT}	VCC ⁽¹⁾	nFAULT	≥ 10 -k Ω pullup
R_{SNSOUT}	VCC ⁽¹⁾	SNSOUT	≥ 10 -k Ω pullup
R_{SENSE}	SP	SN/GND	Optional low-side sense resistor

- (1) VCC is not a pin on the DRV8701, but a VCC supply voltage pullup is required for open-drain outputs nFAULT and SNSOUT. The system controller supply can be used for this pullup voltage, or these pins can be pulled up to either AVDD or DVDD.

External FETs

Component	Gate	Drain	Source	Recommended
Q_{HS1}	GH1	VM	SH1	Supports up to 200-nC FETs at 40-kHz PWM; see Detailed Design Procedure for more details
Q_{LS1}	GL1	SH1	SP or GND	
Q_{HS2}	GH2	VM	SH2	
Q_{LS2}	GL2	SH2	SP or GND	

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range referenced with respect to GND (unless otherwise noted) ⁽¹⁾

	MIN	MAX	UNIT
Power supply voltage (VM)	−0.3	47	V
Power supply voltage ramp rate (VM)	0	2	V/μs
Charge pump voltage (VCP, CPH)	−0.3	VM + 12	V
Charge pump negative switching pin (CPL)	−0.3	VM	V
Internal logic regulator voltage (DVDD)	−0.3	3.8	V
Internal analog regulator voltage (AVDD)	−0.3	5.75	V
Control pin voltage (PH, EN, IN1, IN2, nSLEEP, nFAULT, VREF, IDRIVE, SNSOUT)	−0.3	5.75	V
High-side gate pin voltage (GH1, GH2)	−0.3	VM + 12	V
Continuous phase node pin voltage (SH1, SH2)	−1.2	VM + 1.2	V
Pulsed 10 μs phase node pin voltage (SH1, SH2)	−2.0	VM + 2	V
Low-side gate pin voltage (GL1, GL2)	−0.3	12	V
Continuous shunt amplifier input pin voltage (SP, SN)	−0.5	1	V
Pulsed 10-μs shunt amplifier input pin voltage (SP, SN)	−1	1	V
Shunt amplifier output pin voltage (SO)	−0.3	5.75	V
Open-drain output current (nFAULT, SNSOUT)	0	10	mA
Gate pin source current (GH1, GL1, GH2, GL2)	0	250	mA
Gate pin sink current (GH1, GL1, GH2, GL2)	0	500	mA
Shunt amplifier output pin current (SO)	0	5	mA
Operating junction temperature, T _J	−40	150	°C
Storage temperature, T _{stg}	−65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

6.2 ESD Ratings

	VALUE	UNIT
V _(ESD) Electrostatic discharge	Human body model (HBM) ESD stress voltage ⁽¹⁾	±2000
	Charged device model (CDM) ESD stress voltage ⁽²⁾	±500

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

	MIN	MAX	UNIT
VM Power supply voltage range	5.9	45	V
VCC Logic level input voltage	0	5.5	V
VREF Reference RMS voltage range (VREF)	0.3 ⁽¹⁾	AVDD	V
f _{PWM} Applied PWM signal (PH/EN or IN1/IN2)		100	kHz
I _{AVDD} AVDD external load current		30 ⁽²⁾	mA
I _{DVDD} DVDD external load current		30 ⁽²⁾	mA
I _{SO} Shunt amplifier output current loading (SO)		5	mA
T _A Operating ambient temperature	−40	125	°C

- (1) Operational at VREF = 0 to 0.3 V, but accuracy is degraded
(2) Power dissipation and thermal limits must be observed

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		DRV8701	UNIT
		RGE (VQFN)	
		24 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	34.8	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	37.1	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	12.2	°C/W
ψ_{JT}	Junction-to-top characterization parameter	0.6	°C/W
ψ_{JB}	Junction-to-board characterization parameter	12.2	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	3.7	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.5 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER SUPPLIES (VM, AVDD, DVDD)						
VM	VM operating voltage		5.9		45	V
I _{VM}	VM operating supply current	VM = 24 V; nSLEEP high		6	9.5	mA
I _{VMQ}	VM sleep mode supply current	nSLEEP = 0 VM = 24 V		9 14	15 25	μA
t _{SLEEP}	Sleep time	nSLEEP low to sleep mode			100	μs
t _{WAKE}	Wake-up time	nSLEEP high to output change			1	ms
t _{ON}	Turn-on time	VM > UVLO to output transition			1	ms
DVDD	Internal logic regulator voltage	External load 0 to 30 mA	3.0	3.3	3.5	V
AVDD	Internal logic regulator voltage	External load 0 to 30 mA	4.4	4.8	5.2	V
CHARGE PUMP (VCP, CPH, CPL)						
VCP	VCP operating voltage	VM = 12 V; I _{VCP} = 0 to 12 mA	20.5	21.5	22.5	V
		VM = 8 V; I _{VCP} = 0 to 10 mA	13.5	14.4	15	
		VM = 5.9 V; I _{VCP} = 0 to 8 mA	9.4	9.9	10.4	
I _{VCP}	Charge pump current capacity	VM > 12 V	12			mA
		8 V < VM < 12 V	10			
		5.9 V < VM < 8 V	8			
f _{VCP} ⁽¹⁾	Charge pump switching frequency	VM > UVLO	200	400	700	kHz
CONTROL INPUTS (PH, EN, IN1, IN2, nSLEEP)						
V _{IL}	Input logic low voltage				0.8	V
V _{IH}	Input logic high voltage		1.5			V
V _{HYS}	Input logic hysteresis		100			mV
I _{IL}	Input logic low current	V _{IN} = 0 V	–5		5	μA
I _{IH}	Input logic high current	V _{IN} = 5 V			78	μA
R _{PD}	Pulldown resistance		64	115	173	kΩ
t _{PD}	Propagation delay	PH/EN, IN1/IN2 to GHx/GLx		500		ns
CONTROL OUTPUTS (nFAULT, SNSOUT)						
V _{OL}	Output logic low voltage	I _O = 2 mA			0.1	V
I _{OZ}	Output high impedance leakage	V _{IN} = 5 V	–2		2	μA
FET GATE DRIVERS (GH1, GH2, SH1, SH2, GL1, GL2)						
V _{GHS}	High-side VGS gate drive (gate-to-source)	VM > 12 V; V _{GHS} with respect to SHx	8.5	9.5	10.5	V
		VM = 8 V; V _{GHS} with respect to SHx	5.5	6.4	7	
		VM = 5.9 V; V _{GHS} with respect to SHx	3.5	4.0	4.5	
V _{GLS}	Low-side VGS gate drive (gate-to-source)	VM > 12 V	8.5	9.3	10.5	V
		VM = 5.9 V	3.9	4.3	4.9	
t _{DEAD}	Output dead time	Observed t _{DEAD} depends on IDRIVE setting		380		ns
t _{DRIVE}	Gate drive time			2.5		μs
I _{IDRIVE, SRC}	Peak source current	R _{IDRIVE} < 1 kΩ to GND		6		mA
		R _{IDRIVE} = 33 kΩ ±5% to GND		12.5		
		R _{IDRIVE} = 200 kΩ ±5% to GND, or R _{IDRIVE} < 1 kΩ to AVDD		25		
		R _{IDRIVE} > 500 kΩ ±5% to GND		100		
		R _{IDRIVE} = 68 kΩ ±5% to AVDD		150		

(1) Specified by design and characterization data

Electrical Characteristics (continued)

over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I _{DRIVE,SNK}	Peak sink current	R _{IDRIVE} < 1 kΩ to GND		12.5		mA
		R _{IDRIVE} = 33 kΩ ±5% to GND		25		
		R _{IDRIVE} = 200 kΩ ±5% to GND		50		
		R _{IDRIVE} < 1 kΩ to AVDD		200		
		R _{IDRIVE} > 500 ±5% kΩ to GND		300		
		R _{IDRIVE} = 68 kΩ ±5% to AVDD		50		
		I _{HOLD}	FET holding current	Source current after t _{DRIVE}		
Sink current after t _{DRIVE}				25		
I _{STRONG}	FET hold-off strong pulldown	GHx		490		mA
		GLx		690		
R _{OFF}	FET gate hold-off resistor	Pulldown GHx to SHx		200		kΩ
		Pulldown GLx to GND		150		
CURRENT SHUNT AMPLIFIER AND PWM CURRENT CONTROL (SP, SN, SO, VREF)						
V _{VREF}	VREF input voltage	For current internal chopping	0.3 ⁽²⁾		AVDD	V
A _V	Amplifier gain	50 < V _{SP} < 200 mV; V _{SN} = GND	18	20	22	V/V
		10 < V _{SP} < 50 mV; V _{SN} = GND	16	20	24	
V _{OFF}	SO offset	V _{SP} = V _{SN} = GND		50	250	mV
I _{SP}	SP input current	V _{SP} = 100 mV; V _{SN} = GND		-40		μA
t _{SET} ⁽³⁾	Settling time to ±1%	V _{SP} = V _{SN} = GND to V _{SP} = 100 mV, V _{SN} = GND			1.5	μs
C _{SO} ⁽³⁾	Allowable SO pin capacitance				1	nF
t _{OFF}	PWM current regulation off-time			25		μs
t _{BLANK}	PWM blanking time			2		μs
PROTECTION CIRCUITS						
V _{UVLO}	VM undervoltage lockout	VM falling; UVLO report		5.4	5.8	V
		VM rising; UVLO recovery		5.6	5.9	
V _{UVLO,HYS}	VM undervoltage hysteresis	Rising to falling threshold	100			mV
t _{UVLO}	VM UVLO falling deglitch time	VM falling; UVLO report		10		μs
V _{CPUV}	Charge pump undervoltage	CPUV report		VM + 2.8		V
V _{DS OCP}	Overcurrent protection trip level, VDS of each external FET	High-side FETs: VM – SHx Low-side FETs: SHx – SP	0.8	1		V
V _{SP OCP}	Overcurrent protection trip level, measured by sense amplifier	V _{SP} voltage with respect to GND	0.8	1		V
t _{OCP}	Overcurrent deglitch time			4.5		μs
t _{RETRY}	Overcurrent retry time			3		ms
T _{TSD} ⁽³⁾	Thermal shutdown temperature	Die temperature, T _J	150			°C
T _{HYS} ⁽³⁾	Thermal shutdown hysteresis	Die temperature, T _J		20		°C
V _{GS CLAMP}	Gate drive clamping voltage	Positive clamping voltage	10.5		13	V
		Negative clamping voltage	–1	–0.7	–0.5	

 (2) Operational at $V_{\text{REF}} = 0$ to 0.3 V, but accuracy is degraded

(3) Specified by design and characterization data

6.6 Typical Characteristics

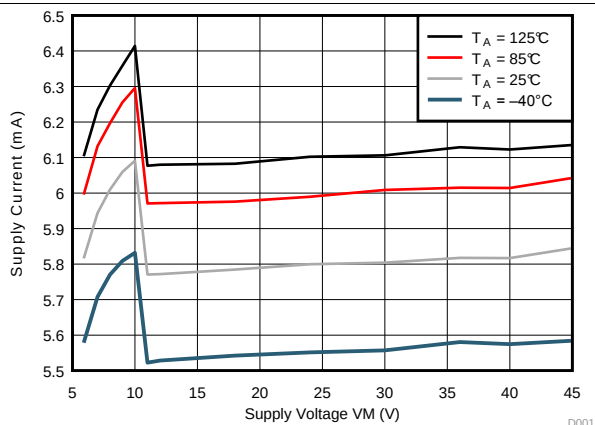


Figure 1. Supply Current over VM

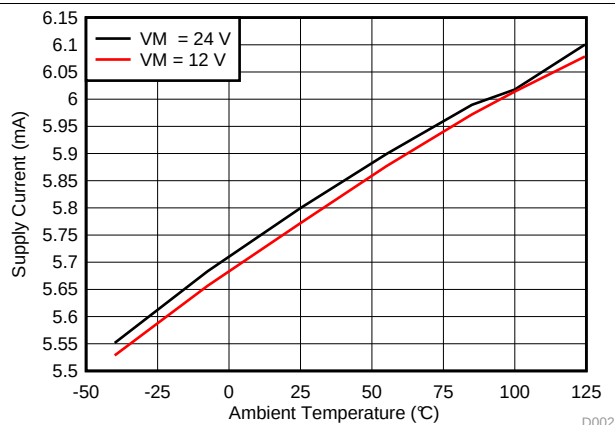


Figure 2. Supply Current over Temperature

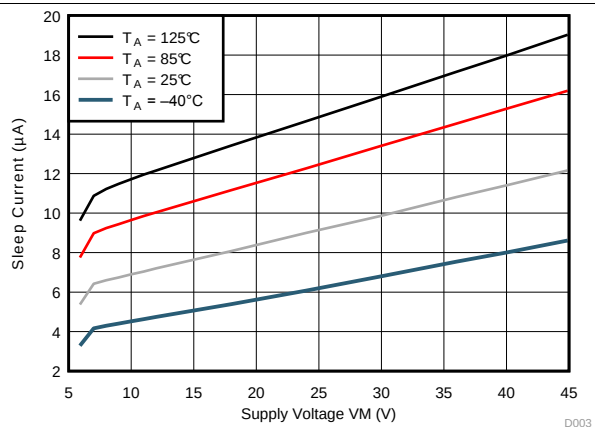


Figure 3. Sleep Current over VM

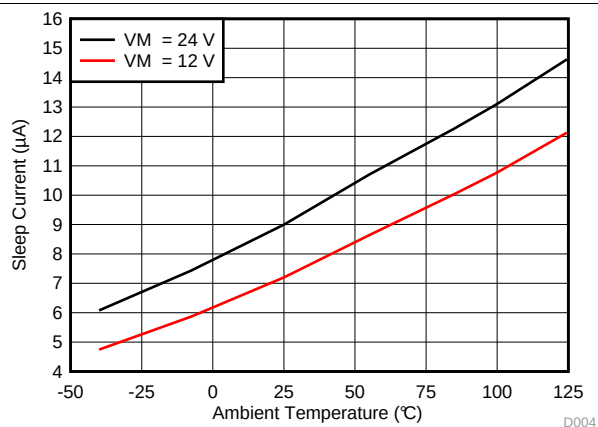


Figure 4. Sleep Current over Temperature

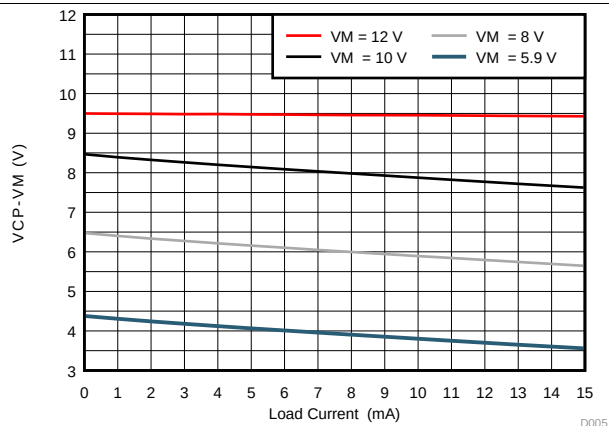


Figure 5. VCP over Load ($T_A = 25^\circ\text{C}$)

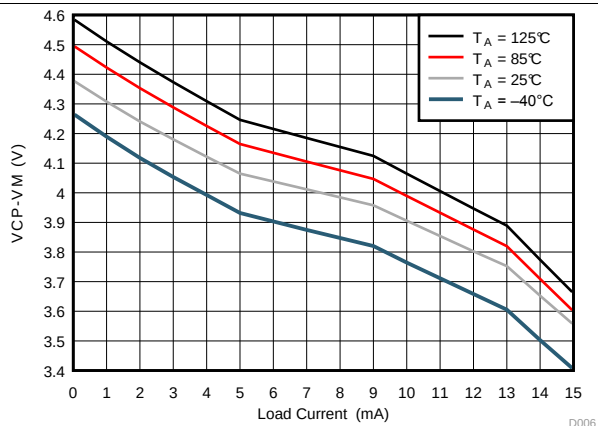


Figure 6. VCP over Load ($V_M = 5.9\text{ V}$)

Typical Characteristics (continued)

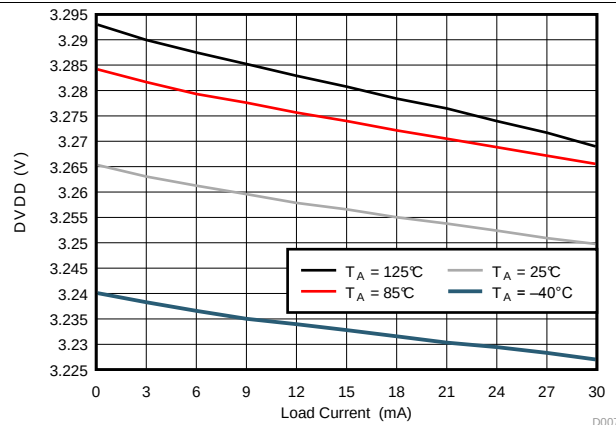


Figure 7. DVDD Regulator over Load (VM = 12 V)

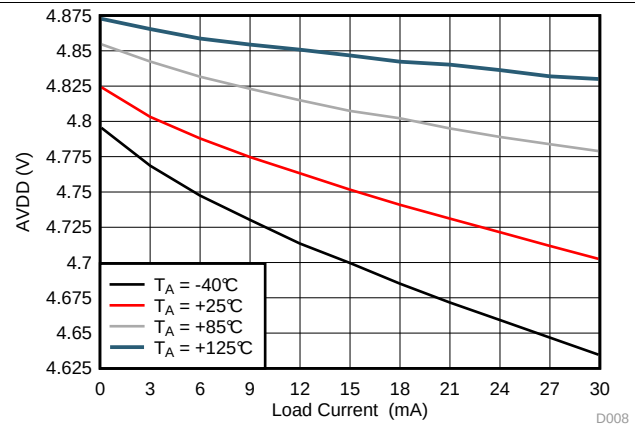


Figure 8. AVDD Regulator over Load (VM = 12 V)

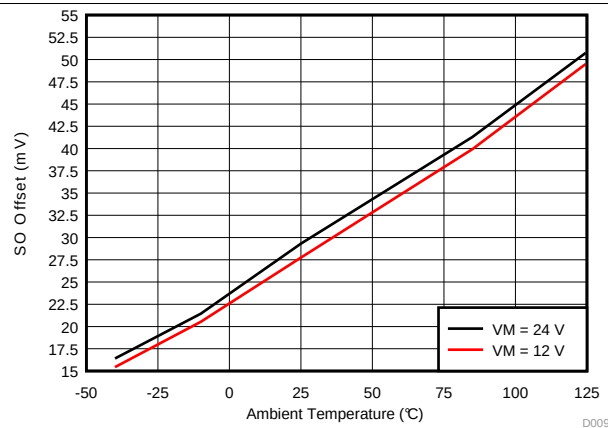


Figure 9. SO Offset over Temperature

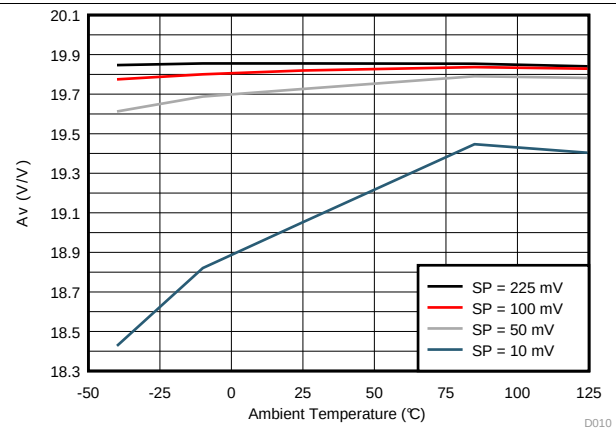


Figure 10. Amplifier Gain over Temperature

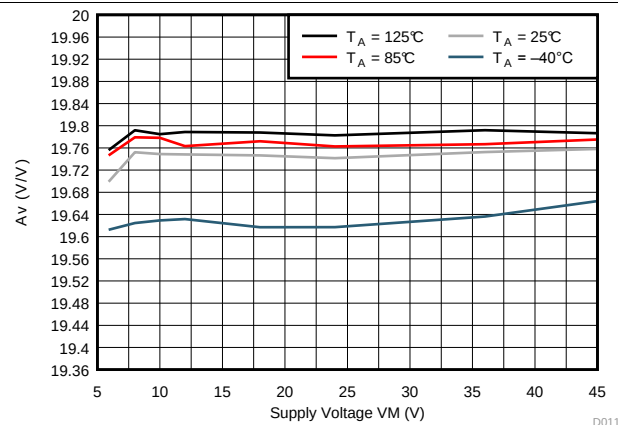


Figure 11. Amplifier Gain over VM (SP = 50 mV)

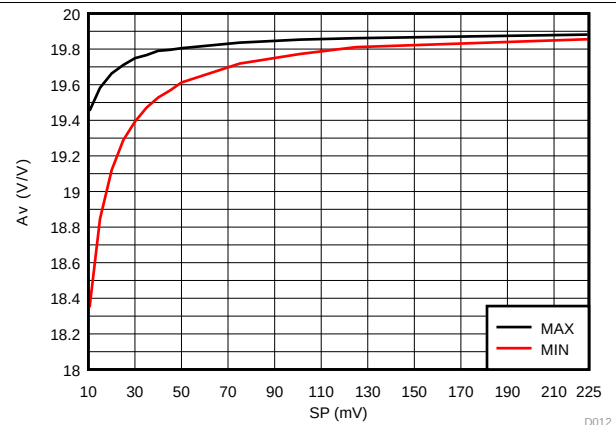


Figure 12. Amplifier Gain over VM and Temperature Range

Typical Characteristics (continued)

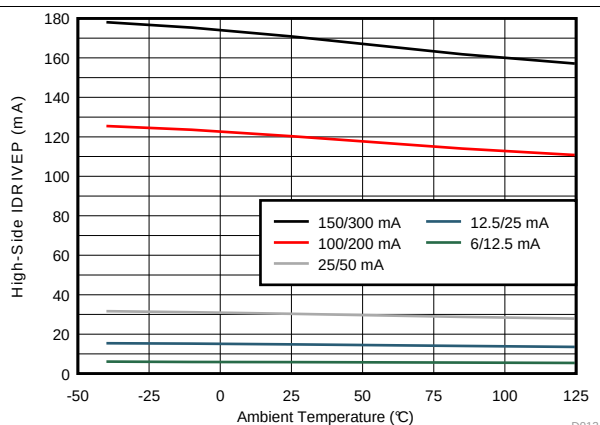


Figure 13. High-Side IDRIVEP over Temperature (VM = 12 V)

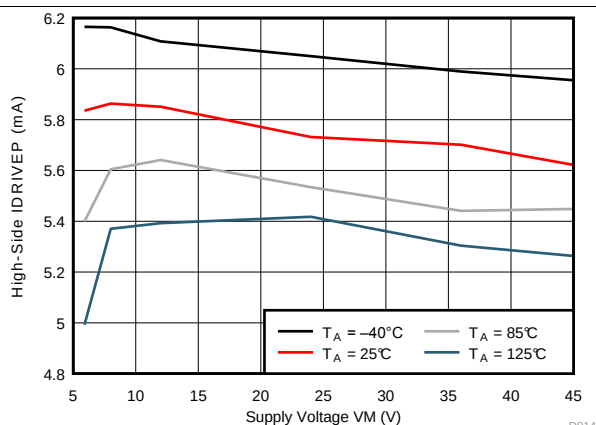


Figure 14. 6-/12.5-mA High-Side IDRIVEP over VM

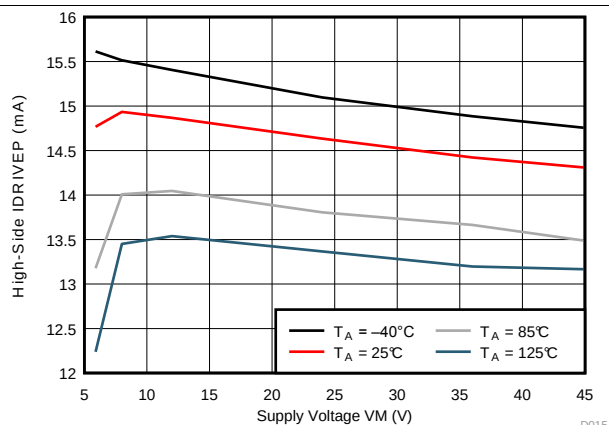


Figure 15. 12.5-/25-mA High-Side IDRIVEP over VM

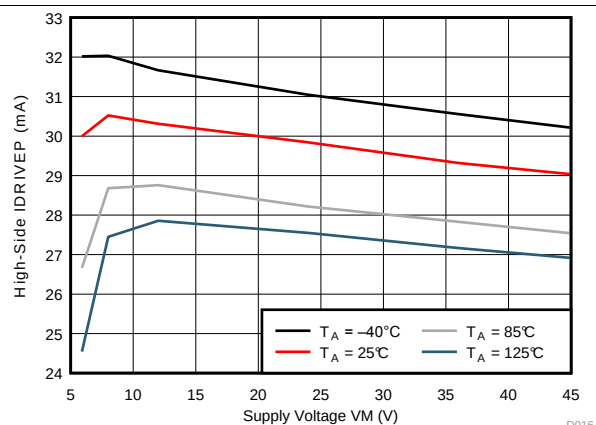


Figure 16. 25-/50-mA High-Side IDRIVEP over VM

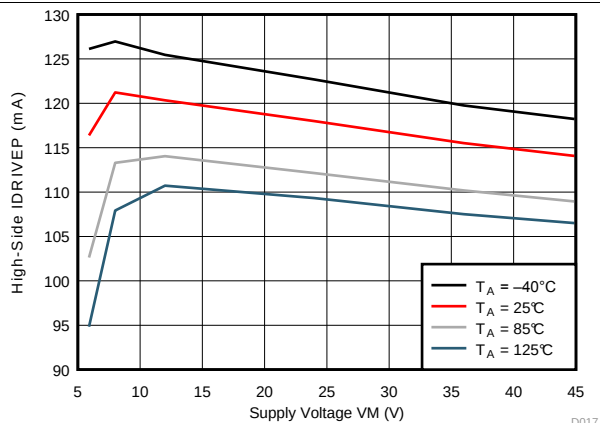


Figure 17. 100-/200-mA High-Side IDRIVEP over VM

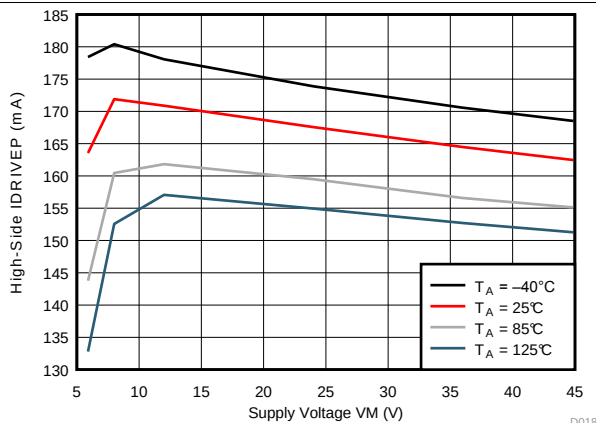


Figure 18. 150-/300-mA High-Side IDRIVEP over VM

7 Detailed Description

7.1 Overview

The DRV8701 is an H-bridge gate driver (also called a pre-driver or controller). The device integrates FET gate drivers in order to control four external NMOS FETs. The device can be powered with a supply voltage between 5.9 and 45 V.

A simple PH/EN (DRV8701E) or PWM (DRV8701P) interface allows interfacing to the controller circuit.

A low-power sleep mode is included, which can be enabled using the nSLEEP pin.

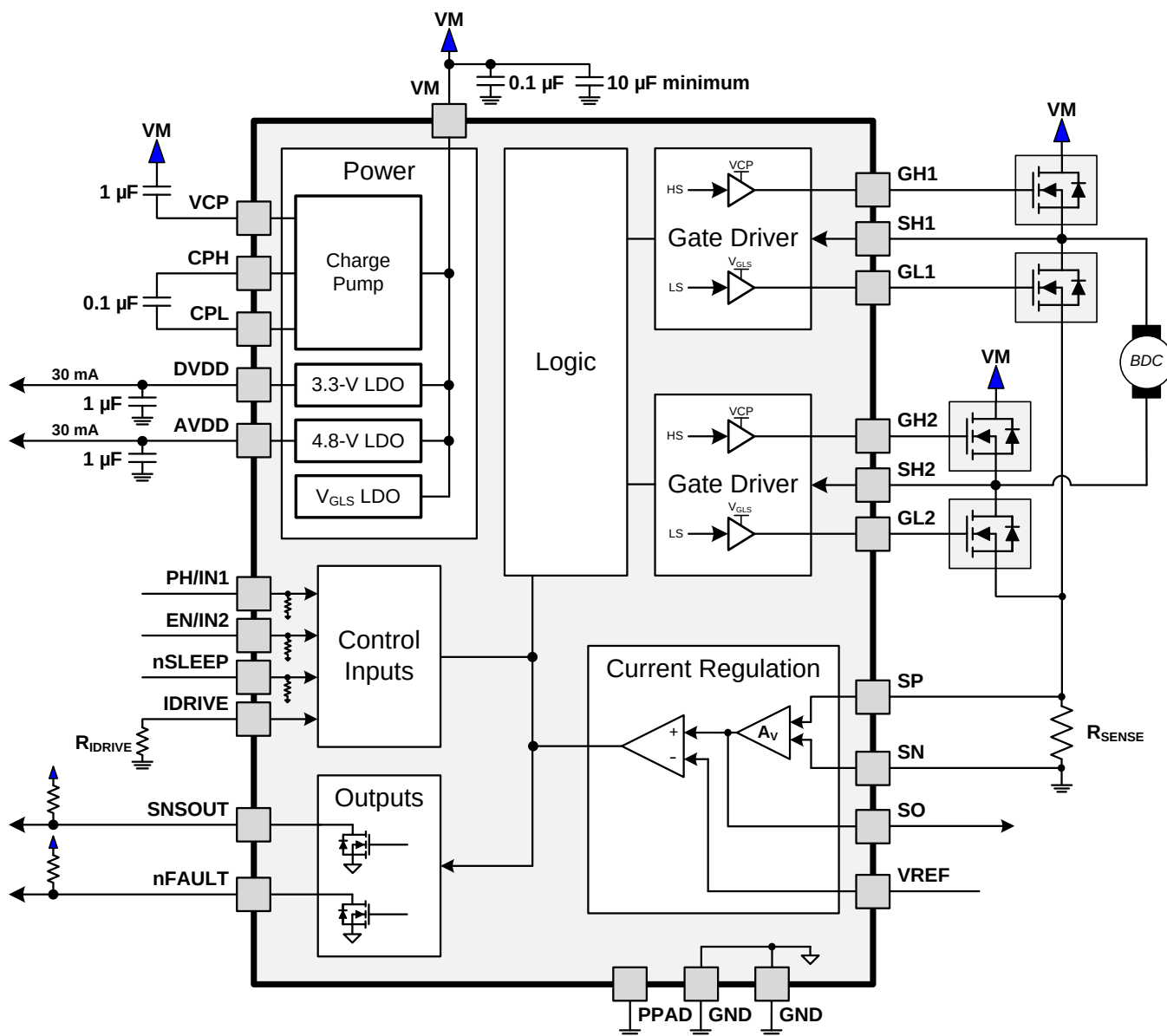
The gate drive strength can be adjusted to optimize a system for a given FET without adding external resistors in series with the FET gates. The IDRIVE pin allows for selection of the peak current driven into the external FET gate. Both the high-side and low-side FETs are driven with a V_{GS} of 9.5 V nominally when $V_M > 12$ V. At lower V_M voltages, the V_{GS} is reduced. The high-side gate drive voltage is generated using a doubler-architecture charge pump that regulates to $V_M + 9.5$ V.

This device greatly reduces the component count of discrete motor driver systems by integrating the necessary FET drive circuitry into a single device. In addition, the DRV8701 adds protection features above traditional discrete implementations: UVLO, OCP, pre-driver faults, and thermal shutdown.

A start-up (inrush) or running current limitation is built in using a fixed time-off current chopping scheme. The chopping current level is set by choosing the sense resistor value and by setting a voltage on the VREF pin.

A shunt amplifier output is provided for accurate current measurements by the system controller. The SO pin outputs a voltage that is 20 times the voltage seen across the sense resistor.

7.2 Functional Block Diagram



7.3 Feature Description

7.3.1 Bridge Control

The DRV8701E is controlled using a PH/EN interface. The following logic table (Table 1) gives the full H-bridge state when driving a single brushed DC motor. Note that Table 1 does not take into account the current control built into the DRV8701E. Positive current is defined in the direction of xOUT1 → xOUT2.

Table 1. DRV8701E (PH/EN) Control Interface

nSLEEP	EN	PH	SH1	SH2	AVDD/DVDD	Description
0	X	X	High-Z	High-Z	Disabled	Sleep mode; H-bridge disabled High-Z
1	0	X	L	L	Enabled	Brake, low-side slow decay
1	1	0	L	H	Enabled	Reverse drive (current SH2 → SH1)
1	1	1	H	L	Enabled	Forward drive (current SH1 → SH2)

The DRV8701P is controlled using a PWM interface (IN1/IN2). The following logic table (Table 2) gives the full H-bridge state when driving a single brushed DC motor. Note that Table 2 does not take into account the current control built into the DRV8701P. Positive current is defined in the direction of xOUT1 → xOUT2.

Table 2. DRV8701P (PWM) Control Interface

nSLEEP	IN1	IN2	SH1	SH2	AVDD/DVDD	Description
0	X	X	High-Z	High-Z	Disabled	Sleep mode; H-bridge disabled High-Z
1	0	0	High-Z	High-Z	Enabled	Coast; H-bridge disabled High-Z
1	0	1	L	H	Enabled	Reverse (current SH2 → SH1)
1	1	0	H	L	Enabled	Forward (current SH1 → SH2)
1	1	1	L	L	Enabled	Brake; low-side slow decay

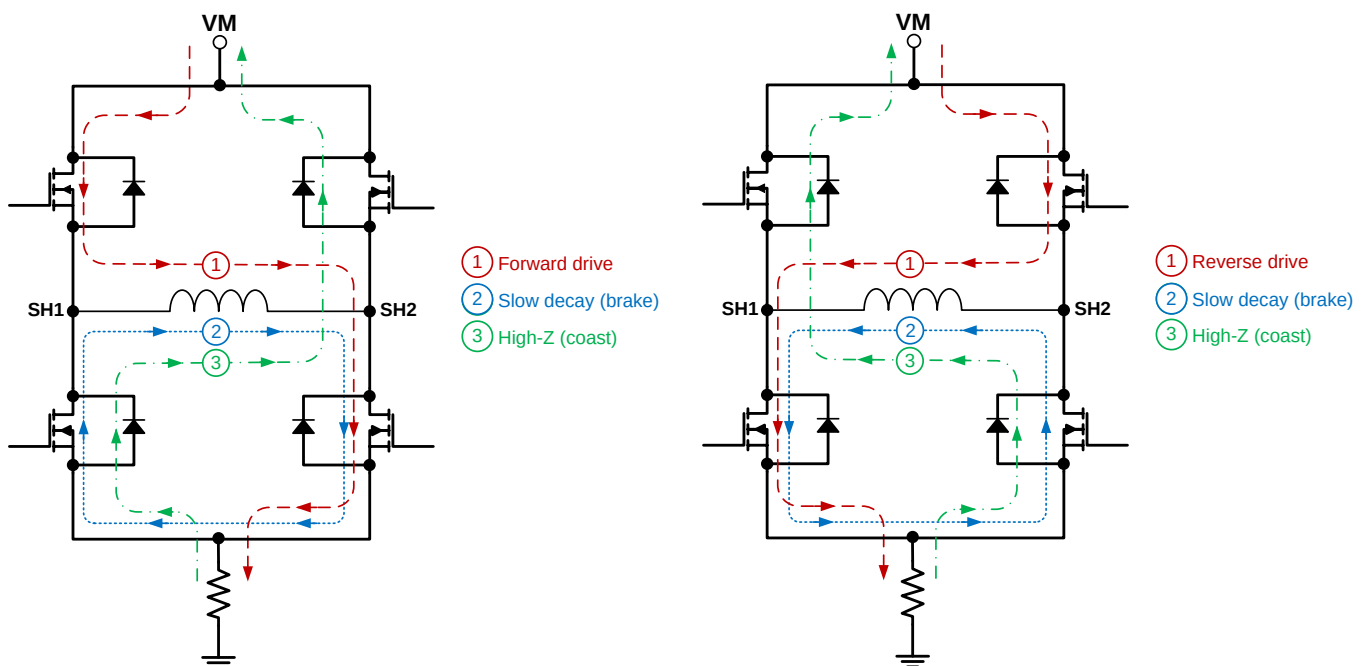


Figure 19. H-Bridge Operational States

7.3.2 Half-Bridge Operation

The DRV8701 can be used to drive only a single half-bridge instead of a full H-bridge. To operate in this mode, leave GH1 and GL1 disconnected. Also, connect a 1/10 W, 330-Ω 5% resistor from SH1 to GND.

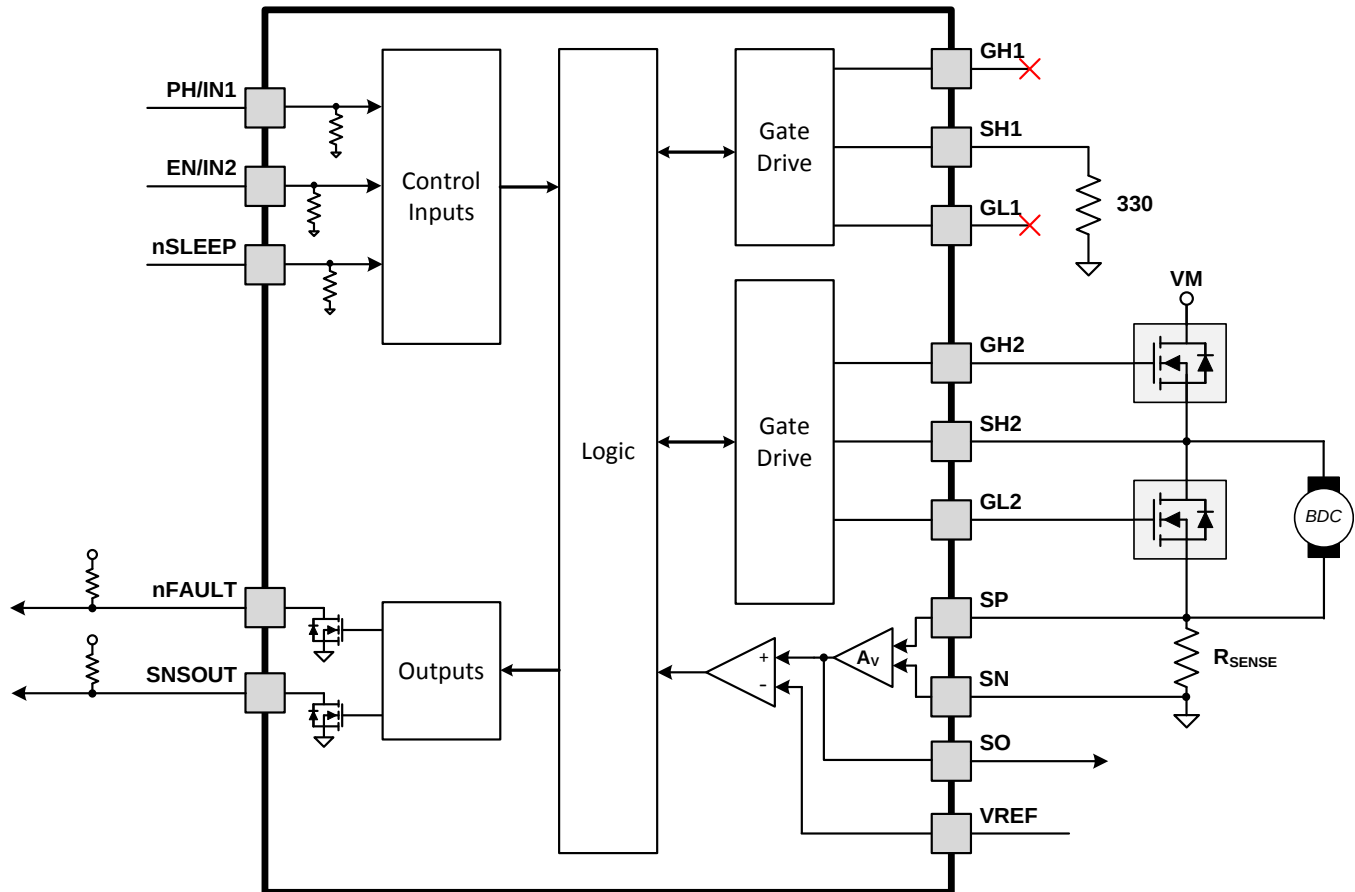


Figure 20. Half-H Bridge Operation Mode

For the DRV8701E, this mode is controlled by tying the PH pin low. Table 3 gives the control scheme. EN = 1 enables the high-side FET, and EN = 0 enables the low-side FET. EN = 1 and PH = 1 is an invalid state.

Table 3. DRV8701E (PH/EN) Control Interface for Half-H Bridge Mode

nSLEEP	EN	PH	SH2	AVDD/DVDD	Description
0	X	X	High-Z	Disabled	Sleep mode; disabled High-Z
1	0	X	L	Enabled	Brake, low-side slow decay
1	1	0	H	Enabled	Drive (Current SH2 → GND)
1	1	1			Invalid state

For the DRV8701P, Table 4 gives the control scheme. IN1 = 1 and IN2 = 0 is an invalid state.

Table 4. DRV8701P (PWM) Control Interface for Half-H Bridge Mode

nSLEEP	IN1	IN2	SH2	AVDD/DVDD	Description
0	X	X	High-Z	Disabled	Sleep mode; disabled High-Z
1	0	0	High-Z	Enabled	Coast; disabled High-Z
1	0	1	H	Enabled	Drive (current SH2 → GND)
1	1	0			Invalid state
1	1	1	L	Enabled	Brake; low-side slow decay

7.3.3 Current Regulation

The maximum current through the motor winding is regulated by a fixed off-time PWM current regulation, or current chopping. When an H-bridge is enabled in forward or reverse drive, current rises through the winding at a rate dependent on the DC voltage and inductance of the winding. After the current hits the current chopping threshold, the bridge enters a brake (low-side slow decay) mode until t_{OFF} has expired.

Note that immediately after the current is enabled, the voltage on the SP pin is ignored for a period of time (t_{BLANK}) before enabling the current sense circuitry.

The PWM chopping current is set by a comparator which compares the voltage across a current sense resistor connected to the SP pin, multiplied by a factor of A_V , with a reference voltage from the VREF pin. The factor A_V is the shunt amplifier gain, which is 20 V/V in the DRV8701.

The chopping current is calculated as follows:

$$I_{CHOP} = \frac{V_{REF} - V_{OFF}}{A_V \times R_{SENSE}} \quad (1)$$

Example: If a 50 mΩ sense resistor is used and $V_{REF} = 3.3$ V, the full-scale chopping current will be 3.25 A. A_V is 20 V/V and V_{OFF} is assumed to be 50 mV in this example.

For DC motors, current regulation is generally used to limit the start-up and stall current of the motor. If the current regulation feature is not needed, it can be disabled by tying VREF directly to AVDD and tying SP and SN to GND.

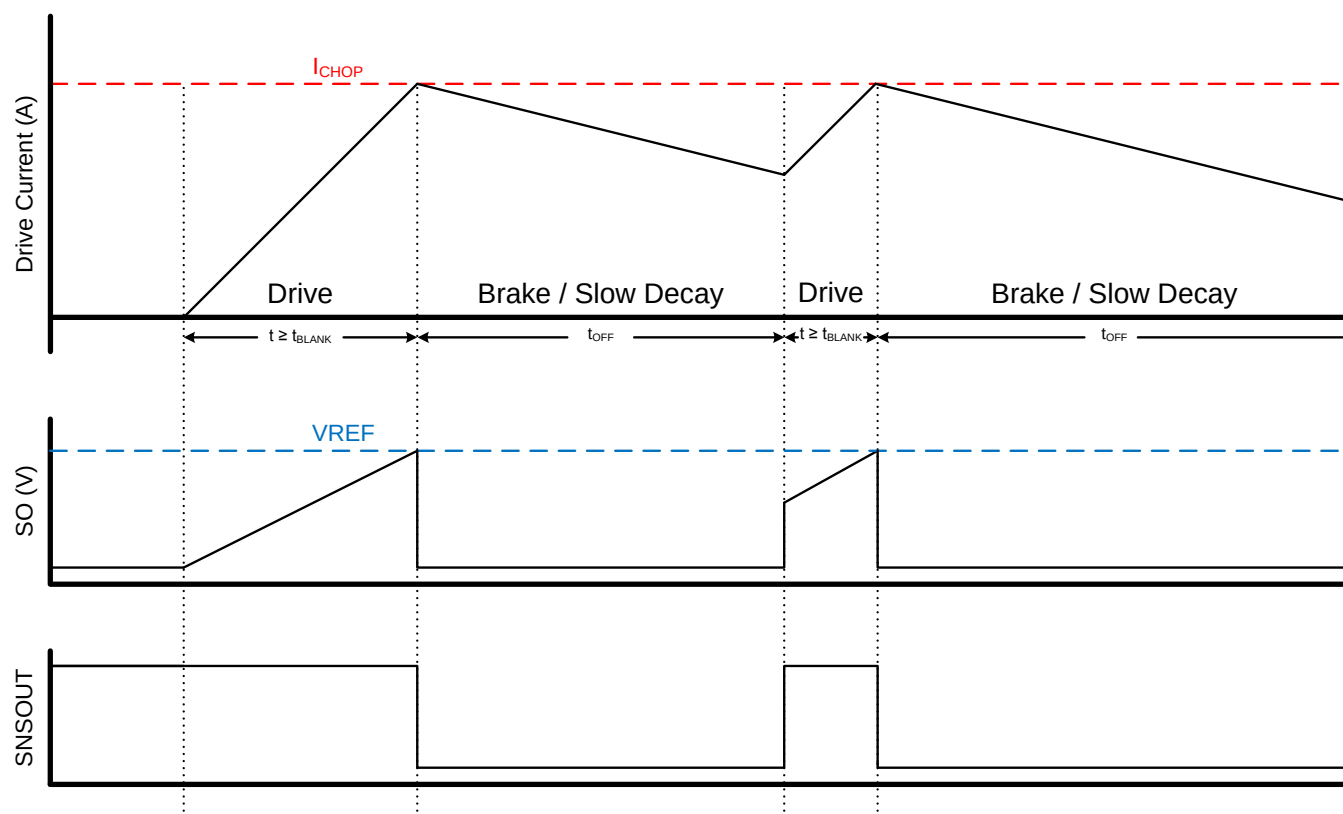


Figure 21. Sense Amplifier and Current Chopping Operation

During brake mode (slow decay), current is recirculated through the low-side FETs. Because current is not flowing through the sense resistor, SO does not represent the motor current.

7.3.4 Amplifier Output SO

The SO pin on the DRV8701 outputs an analog voltage equal to the voltage seen across the SP and SN pins multiplied by A_V . The factor A_V is the shunt amplifier gain, which is 20 V/V in the DRV8701. SO is only valid during forward or reverse drive. The H-bridge current is approximately equal to:

$$I = \frac{SO - V_{OFF}}{A_V \times R_{SENSE}} \quad (2)$$

When SP and SN are 0 V, SO outputs the amplifier offset voltage V_{OFF} . No capacitor is required on the SO pin.

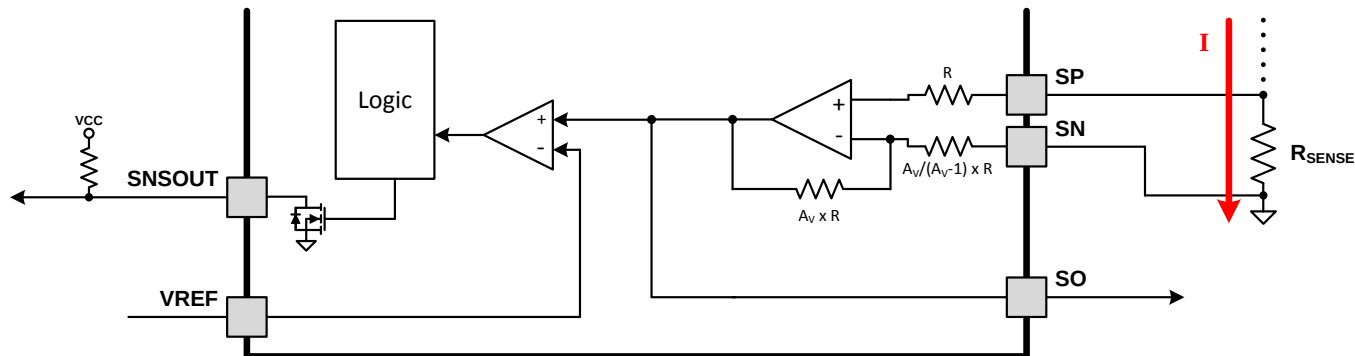


Figure 22. Sense Amplifier Diagram

If the voltage across SP and SN exceeds 1 V, then the DRV8701 flags an overcurrent condition.

The SO pin can source up to 5 mA of current. If the pin is shorted to GND, or if a higher-current load is driven by this pin, the output acts as a constant-current source. The output voltage is not representative of the H-bridge current in this state.

This shunt amplifier feature can be disabled by tying the SP and SN pins to GND. When the amplifier is disabled, current regulation is also disabled.

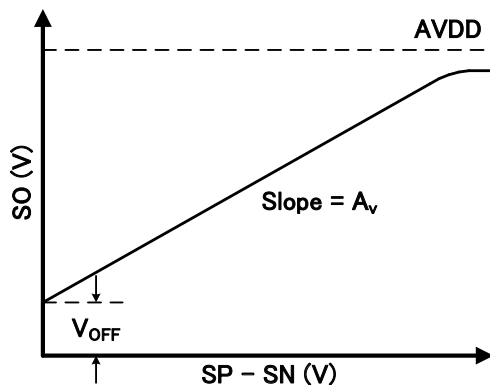


Figure 23. Sense Amplifier Output

7.3.4.1 SNSOUT

The SNSOUT pin of the DRV8701 indicates when the device is in current chopping mode. When the driver is in a slow decay mode caused by internal PWM current chopping (I_{CHOP} threshold hit), the open-drain SNSOUT output is pulled low. If the current regulation is disabled, then the SNSOUT pin will be high-Z.

Note that if the H-bridge is put into a slow decay mode using the inputs (PH/EN or IN1/IN2), then SNSOUT is not pulled low.

During forward or reverse drive mode, SNSOUT is high until the DRV8701 is internally forced into current chopping. If the drive current rises above I_{CHOP} , the driver enters a brake mode (low-side slow decay). The SNSOUT pin will be pulled low during this current chopping brake mode. After the driver is re-enabled, the SNSOUT pin is released high-Z and the drive mode is restarted.

7.3.5 PWM Motor Gate Drivers

The DRV8701 contains gate drivers for a single H-bridge with external NMOS FETs. Figure 24 shows a block diagram of the gate driver circuitry.

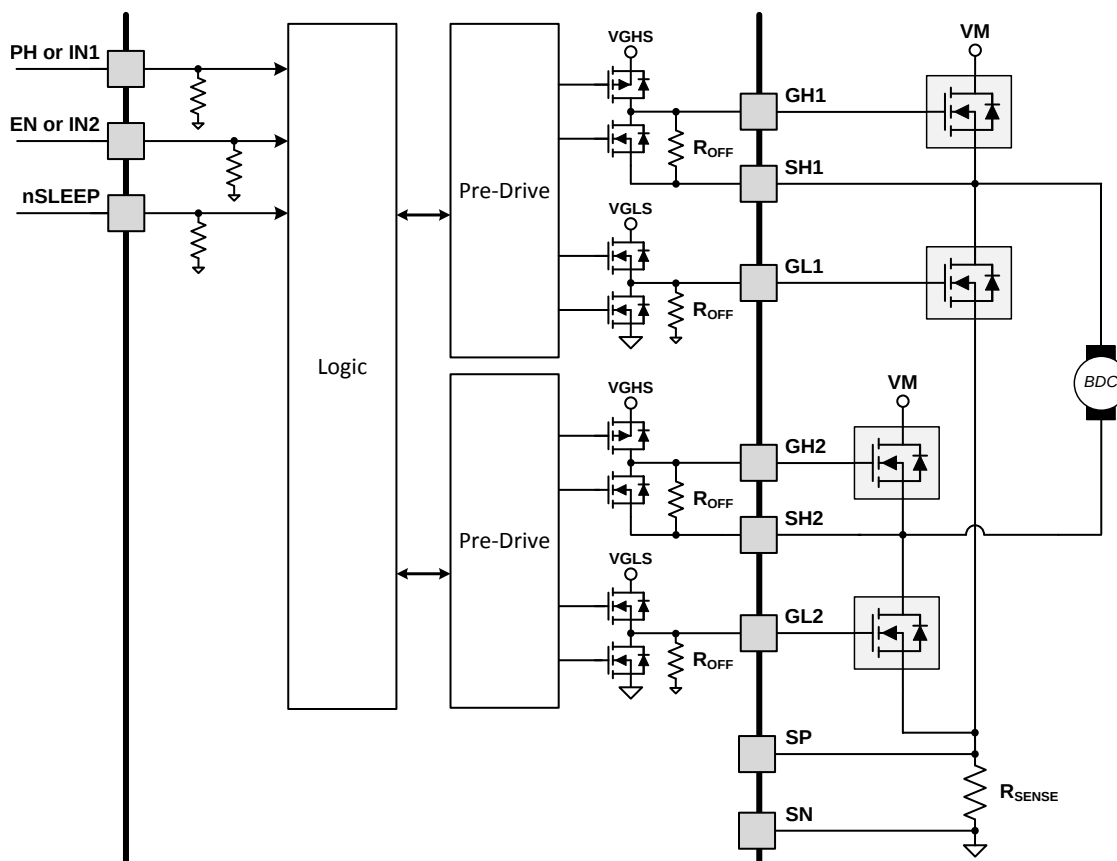


Figure 24. PWM Motor Gate Drivers

Gate drivers inside the DRV8701 directly drive N-channel MOSFETs, which drive the motor current. The high-side gate drive is supplied by the charge pump, while the low-side gate drive voltage is generated by an internal regulator.

The peak drive current of the gate drivers is adjustable through the IDRIVE pin. Peak source currents may be set to 6, 12.5, 25, 100, or 150 mA. The peak sink current is approximately 2× the peak source current. Adjusting the peak current changes the output slew rate, which also depends on the FET input capacitance and gate charge.

The peak drive current is selected by setting the value of the R_{IDRIVE} resistor on the IDRIVE pin or by forcing a voltage onto the IDRIVE pin (see Table 6 for details).

Fast switching times can cause extra voltage noise on VM and GND. This can be especially due to a relatively slow reverse-recovery time of the low-side body diode, where it conducts reverse-bias momentarily, being similar to shoot-through. Slow switching times can cause excessive power dissipation since the external FETs take a longer time to turn on and turn off.

When changing the state of the output, the peak current (I_{DRIVE}) is applied for a short drive period (t_{DRIVE}) to charge the gate capacitance. After this time, a weaker current source (I_{HOLD}) is used to keep the gate at the desired state. When selecting the gate drive strength for a given external FET, the selected current must be high enough to fully charge and discharge the gate during t_{DRIVE} , or excessive power will be dissipated in the FET.

During high-side turn-on, the low-side gate is pulled low with a strong pull-down (I_{STRONG}). This prevents the low-side FET Q_{GS} from charging and keeps the FET off, even when there is fast switching at the outputs.

The pre-driver circuits include enforcement of a dead time in analog circuitry, which prevents the high-side and low-side FETs from conducting at the same time. When switching FETs on, this handshaking prevents the high- or low-side FET from turning on until the opposite FET has been turned off.

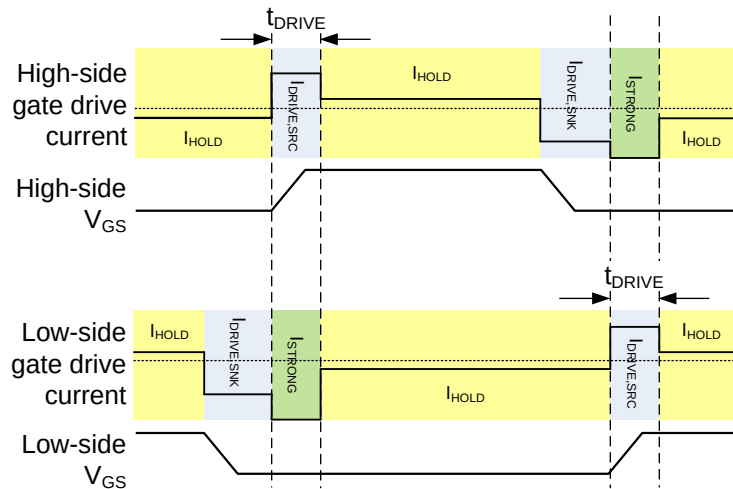


Figure 25. Gate Driver Output to Control External FETs

Q_{GD} Miller charge

When a FET gate is turned on, three different capacitances must be charged.

- Q_{GS} – Gate-to-source charge
- Q_{GD} – Gate-to-drain charge (miller charge)
- Remaining Q_{G}

The FET output is slewing primarily during the Q_{GD} charge.

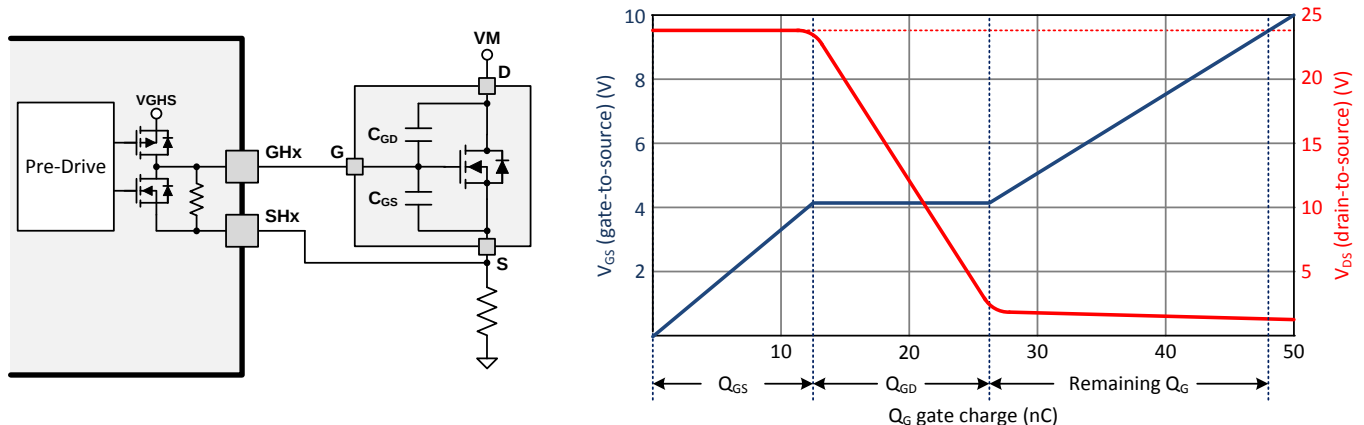


Figure 26. Example FET Gate Charging Profile

7.3.6 IDRIVE Pin

The rise and fall times of the H-bridge output (SHx pins) can be adjusted by setting the IDRIVE resistor value or forcing a voltage onto the IDRIVE pin. The FET gate voltage ramps faster if a higher IDRIVE setting is chosen. The FET gate ramp directly affects the H-bridge output rise and fall time.

Tying IDRIVE to GND selects the lowest drive setting of 6-mA source and 12.5-mA sink. If this pin is left unconnected, then the 100-mA source and 200-mA sink setting are selected.

If IDRIVE is shorted to AVDD, then the VDS OCP monitor on the high-side FETs is disabled. In this setting, the gate driver is configured as 25-mA source and 50-mA sink.

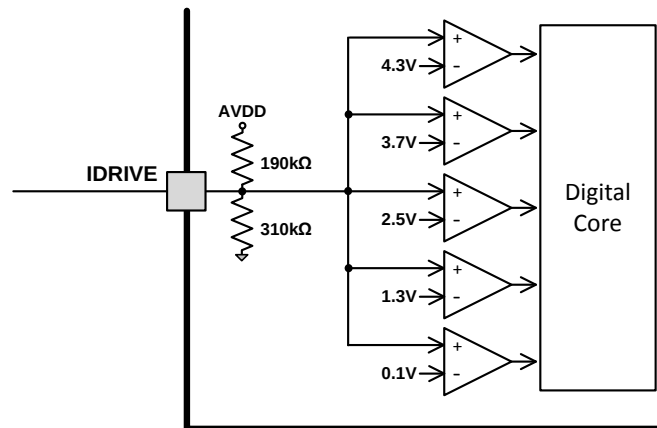
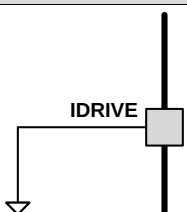
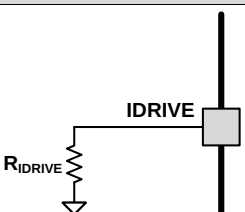
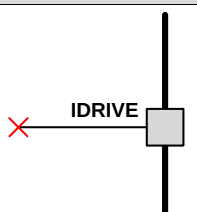
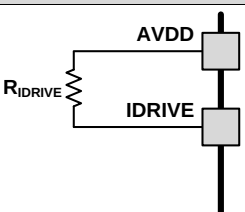
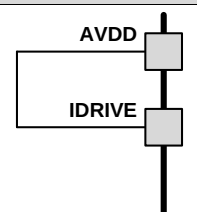


Figure 27. IDRIVE Pin Internal Circuitry

Table 5. IDRIVE Pin Configuration Settings

IDRIVE Resistance	IDRIVE Voltage	Source Current (I _{IDRIVE, SRC})	Sink Current (I _{IDRIVE, SNK})	HS OCP Monitor
<1 kΩ to GND	GND	6 mA	12.5 mA	ON
33 kΩ ±5% to GND	0.7 V ±5%	12.5 mA	25 mA	ON
200 kΩ ±5% to GND	2 V ±5%	25 mA	50 mA	ON
>500 kΩ to GND, High-Z	3 V ±5%	100 mA	200 mA	ON
68 kΩ ±5% to AVDD	4 V ±5%	150 mA	300 mA	ON
<1 kΩ to AVDD	AVDD	25 mA	50 mA	OFF

Table 6. IDRIVE Pin Resistor Settings

<1 kΩ to GND	33 kΩ ±5% to GND 200 kΩ ±5% to GND	>500 kΩ to GND, High-Z	68 kΩ ±5% to AVDD	<1 kΩ to AVDD
				
IDRIVE	IDRIVE	IDRIVE	IDRIVE	IDRIVE
6 / 12.5 mA	12.5 / 25 mA (33 kΩ) 25 / 50 mA (200 kΩ)	100 / 200 mA	150 / 300 mA	25 / 50 mA HS OCP monitor off

7.3.7 Dead Time

Dead time (t_{DEAD}) is measured as the time when SHx is High-Z between turning off one of the H-bridge FETs and turning on the other. For example, the output is High-Z between turning off the high-side FET and turning on the low-side FET.

The DRV8701 inserts a digital dead time of approximately 150 ns. The total dead time also includes the FET gate turn-on time.

The total dead time is dependent on the IDRIVE resistor setting because a portion of the FET gate ramp (GHx and GLx pins) includes the observable dead time.

7.3.8 Propagation Delay

The propagation delay time (t_{DELAY}) is measured as the time between an input edge to an output change. This time is composed of two parts: an input deglitch time and output slewing delay. The input deglitcher prevents noise on the input pins from affecting the output state.

The gate drive slew rate also contributes to the delay time. For the output to change state during normal operation, first, one FET must be turned off. The FET gate is ramped down according to the IDRIVE setting, and the observed propagation delay ends when the FET gate has fallen below the threshold voltage.

7.3.9 Overcurrent V_{DS} Monitor

The gate driver circuit monitors the V_{DS} voltage of each external FET when it is driving current. When the voltage monitored is greater than the OCP threshold voltage ($V_{\text{DS OCP}}$), after the OCP deglitch time (t_{OCP}) has expired, an OCP condition will be detected.

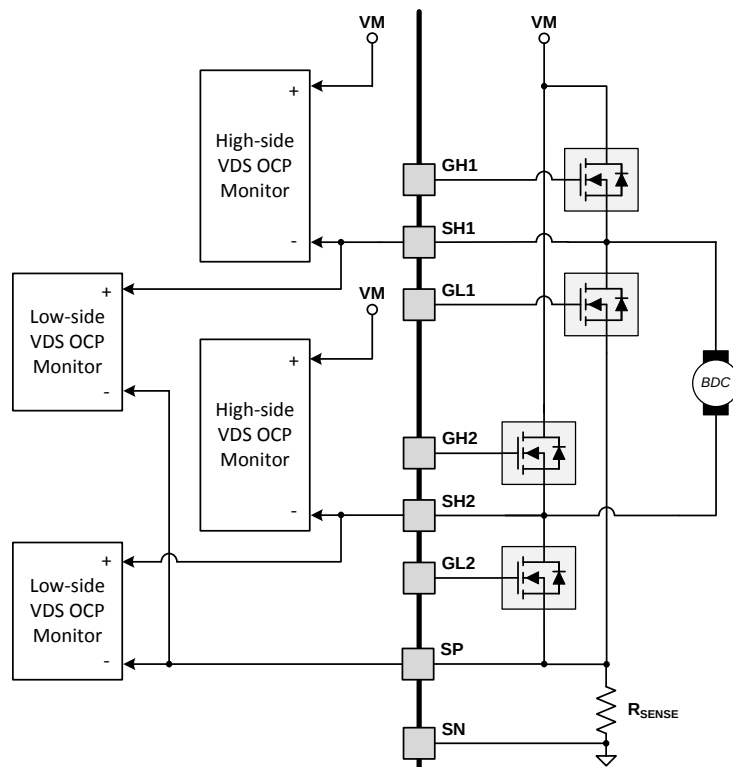


Figure 28. Overcurrent V_{DS} Monitors

When IDRIVE is shorted to AVDD, the V_{DS} OCP monitor on the high-side FETs is disabled. In cases where the VM supplied to the DRV8701 can be different from the external H-bridge supply, this setting must be used in order to prevent false overcurrent detection. In this mode, the IDRIVE current is set to 25-mA source and 50-mA sink.

7.3.10 Charge Pump

A charge pump is integrated to supply a high-side NMOS gate drive voltage of V_{HGS} . The charge pump requires a capacitor between the VM and VCP pins. Additionally a low-ESR ceramic capacitor is required between pins CPH and CPL. When VM is below 12 V, this charge pump behaves as a doubler and generates $VCP = 2 \times VM - 1.5$ V if unloaded. Above $VM = 12$ V, the charge pump regulates VCP such that $VCP = VM + 9.5$ V.

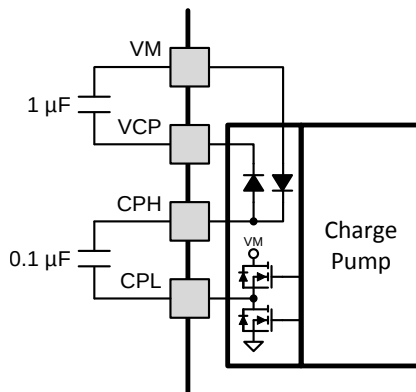


Figure 29. Charge Pump Diagram

7.3.11 LDO Voltage Regulators

Two LDO regulators are integrated into the DRV8701. They can be used to provide the supply voltage for a low-power microcontroller or other low-current devices. For proper operation, bypass the AVDD and DVDD pins to GND using ceramic capacitors.

The AVDD output voltage is nominally 4.8 V, and the DVDD output is nominally 3.3 V. When the AVDD or DVDD current load exceeds 30 mA, the LDO behaves like a constant current source. The output voltage drops significantly with currents greater than this limit.

Note that AVDD and DVDD are disabled when the device is in sleep mode ($nSLEEP = 0$). In addition, when an overtemperature (TSD) or undervoltage (UVLO) fault is encountered, the AVDD regulator is shut off.

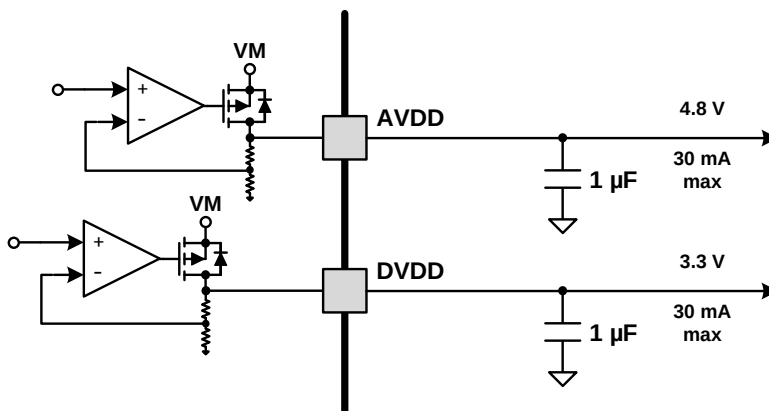


Figure 30. AVDD and DVDD LDOs

The power dissipated in the DRV8701 due to these LDOs may be approximated by:

$$\text{Power} = (VM - AVDD) \times I_{AVDD} + (VM - DVDD) \times I_{DVDD} \quad (3)$$

For example at $VM = 24$ V, drawing 10 mA out of both AVDD and DVDD results in a power dissipation of:

$$\text{Power} = (24 \text{ V} - 4.8 \text{ V}) \times 10 \text{ mA} + (24 \text{ V} - 3.3 \text{ V}) \times 10 \text{ mA} = 192 \text{ mW} + 207 \text{ mW} = 399 \text{ mW} \quad (4)$$

7.3.12 Gate Drive Clamp

A clamping structure limits the gate drive output voltage to V_{GS_CLAMP} to protect the power FETs from damage. The positive voltage clamp is realized using a series of diodes. The negative voltage clamp uses the body diodes of the internal gate driver FET.

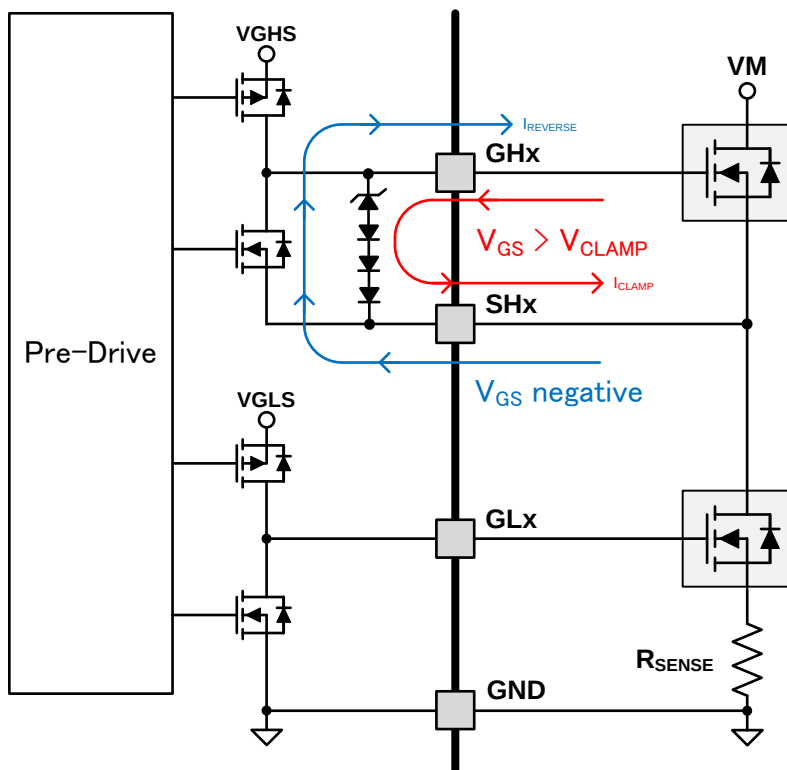


Figure 31. Gate Drive Clamp Diagram

7.3.13 Protection Circuits

The DRV8701 is fully protected against VM undervoltage, charge pump undervoltage, overcurrent, gate driver shorts, and overtemperature events.

7.3.13.1 VM Undervoltage Lockout (UVLO)

If at any time the voltage on the VM pin falls below the UVLO threshold voltage, all FETs in the H-bridge are disabled, the charge pump is disabled, AVDD is disabled, and the nFAULT pin is driven low. Operation resumes when VM rises above the UVLO threshold. The nFAULT pin is released after operation has resumed.

7.3.13.2 VCP Undervoltage Lockout (CPUV)

If at any time the voltage on the VCP pin falls below the charge pump undervoltage threshold voltage (V_{CPUV}), all FETs in the H-bridge are disabled and the nFAULT pin is driven low. Operation resumes when VCP rises above the CPUV threshold. The nFAULT pin is released after operation has resumed.

7.3.13.3 Overcurrent Protection (OCP)

Overcurrent is sensed by monitoring the V_{DS} voltage drop across the external FETs (see Figure 28). If the voltage across a driven FET exceeds the overcurrent trip threshold (V_{DS_OCP}) for longer than the OCP deglitch time (t_{OCP}), an OCP event is recognized. As a result, all FETs in the H-bridge are disabled and the nFAULT pin is driven low; the driver is re-enabled after the OCP retry period (t_{RETRY}) has passed. nFAULT releases high-Z again at after the retry time. If the fault condition is still present, the cycle repeats. If the fault is no longer present, normal operation resumes and nFAULT remains released high-Z.

This V_{DS} overcurrent monitor on the high-side FETs can be disabled by using a specific IDRIVE setting. This allows the system to have a higher DRV8701 VM supply than the H-bridge supply.

In addition to this FET V_{DS} monitor, an overcurrent condition is also detected if the voltage at SP exceeds V_{SP_OCP} .

7.3.13.4 Pre-Driver Fault (PDF)

The GHx and GLx pins are monitored such that if the voltage on the external FET gate does not increase above 1 V (when sourcing current) or decrease below 1 V (when sinking current) after t_{DRIVE} , a pre-driver fault is detected. The device encounters this fault if GHx or GLx are shorted to GND, SHx, or VM. Additionally, the device encounters the pre-driver fault if the IDRIVE setting selected is not sufficient to turn on the external FET. As a result, all FETs in the H-bridge are disabled and the nFAULT pin is driven low. The driver is re-enabled after the retry period (t_{RETRY}) has passed. The nFAULT pin is released after operation has resumed.

7.3.13.5 Thermal Shutdown (TSD)

If the die temperature exceeds T_{TSD} , all FETs in the H-bridge are disabled, the charge pump is shut down, AVDD is disabled, and the nFAULT pin is driven low. After the die temperature has fallen below $T_{TSD} - T_{HYS}$, operation automatically resumes. The nFAULT pin is released after operation has resumed.

Table 7. Fault Response

Fault	Condition	H-Bridge	Charge Pump	AVDD	DVDD	Recovery
VM undervoltage (UVLO)	$VM \leq V_{UVLO}$	Disabled	Disabled	Disabled	Operating	$VM \geq V_{UVLO}$
VCP undervoltage (CPUV)	$VCP < V_{CPUV}$	Disabled	Operating	Operating	Operating	$VCP > V_{CPUV}$
External FET overload (OCP)	$V_{DS} \geq 1.0\text{ V}$ or $V_{SP} - V_{SN} > 1.0\text{ V}$	Disabled	Operating	Operating	Operating	t_{RETRY}
Pre-driver fault (PDF)	Gate voltage unchanged after t_{DRIVE}	Disabled	Operating	Operating	Operating	t_{RETRY}
Thermal shutdown (TSD)	$T_J \geq 150^\circ\text{C}$	Disabled	Disabled	Disabled	Operating	$T_J \leq 130^\circ\text{C}$

7.3.14 Reverse Supply Protection

The following circuit may be implemented to protect the system from reverse supply conditions. This circuit requires the following additional components:

- NMOS FET
- npn BJT
- Diode
- 10-k Ω resistor
- 43-k Ω resistor

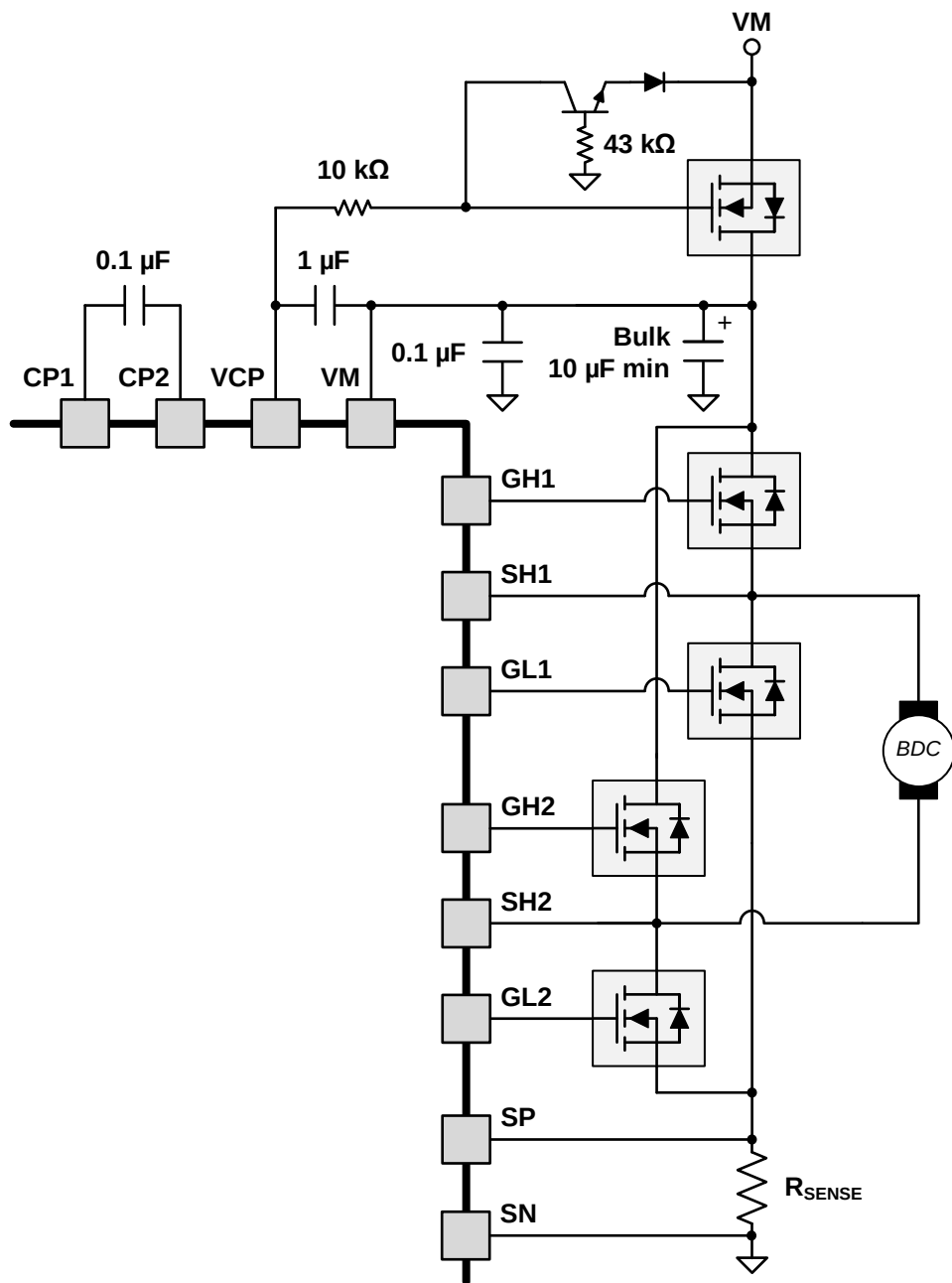


Figure 32. Reverse Supply Protection External Circuitry

7.4 Device Functional Modes

The DRV8701 is active unless the nSLEEP pin is brought low. In sleep mode, the charge pump is disabled, the H-bridge FETs are High-Z, and the AVDD and DVDD regulators are disabled. Note that t_{SLEEP} must elapse after a falling edge on the nSLEEP pin before the device is in sleep mode. The DRV8701 is brought out of sleep mode if nSLEEP is brought high. Note that t_{WAKE} must elapse before the outputs change state after wake-up.

While nSLEEP is brought low, all external H-bridge FETs are disabled. The high-side gate pins GHx are pulled to the output node SHx by an internal resistor, and the low-side gate pins GLx are pulled to GND.

When VM is not applied, and during the power-on time (t_{ON}), the outputs are disabled using weak pulldown resistors between the GHx and SHx pins and between GLx and GND.

Table 8. Functional Modes

Condition		Charge Pump	GHx	GLx	AVDD and DVDD
Unpowered	$VM < V_{\text{UVLO}}$	Disabled	Weak pulldown to SHx	Weak pulldown to GND	Disabled
Sleep mode	$V_{\text{UVLO}} < VM$ nSLEEP low	Disabled	Strong pulldown to GND	Strong pulldown to GND	Disabled
Operating	$V_{\text{UVLO}} < VM$ nSLEEP high	Enabled	Depends on inputs	Depends on inputs	Operating

7.4.1 Operating DRV8701 and H-Bridge on Separate Supplies

The DRV8701 can operate with a different supply voltage (VM) than the system H-bridge supply (VBAT). Case 1 describes normal operation when VM and VBAT are roughly the same. Special considerations must be taken into account for Cases 2, 3, and 4.

- **Case 1:** VM ≈ VBAT. Recommended operation
- **Case 2:** VM > VBAT. IDRIVE must be shorted to AVDD to disable the high-side OCP. The IDRIVE current is fixed at 25-mA source and 50-mA sink. This case can allow the driver to better enhance the external FETs for VBAT < 11.5 V, or operate down to a lower supply voltage below 5.9 V.
- **Case 3:** VM > VBAT (higher than Case 2). IDRIVE must be shorted to AVDD to disable the high-side OCP. This case can also allow the driver to better enhance the external FETs, or operate down to a lower supply voltage below 5.9 V. The IDRIVE current is fixed at 25-mA source and 50-mA sink. Excess gate drive current may be driven through the DRV8701 gate clamps causing additional power dissipation in the DRV8701.
- **Case 4:** VM < VBAT. The high-side FETs may not be in saturation. There may be a significant voltage drop across the high-side FET when driving current. This causes high power dissipation in the external FET. When operating in Case 4, the external FET threshold voltage must be greater than 2 V. Otherwise the DRV8701 will report a pre-driver fault whenever the FET is out of saturation.

Table 9. VM Operational Range based on VBAT

VBAT Range	Case 3	Case 2	Case 1	Case 4
1 V ≤ VBAT < 5.9 V	$VM \geq 0.5 \times VBAT + 5.75 \text{ V}$ $VM \leq 45 \text{ V}$	$VM \geq 5.9 \text{ V}$ $VM < 0.5 \times VBAT + 5.75 \text{ V}$	N/A	N/A
5.9 V ≤ VBAT < 6.4 V		$VM > VBAT$ $VM < 0.5 \times VBAT + 5.75 \text{ V}$	VM = VBAT	$VM \geq 5.9 \text{ V}$ $VM < VBAT$
6.4 V ≤ VBAT < 11.5 V			$VM > 0.6 \times VBAT + 2.5 \text{ V}$ $VM \leq VBAT$	$VM \geq 5.9 \text{ V}$ $VM \leq 0.6 \times VBAT + 2.5 \text{ V}$
11.5 V ≤ VBAT < 14 V	$VM > VBAT$ $VM \leq 45 \text{ V}$	N/A	$VM > VBAT - 4 \text{ V}$ $VM \leq VBAT$	$VM \geq 5.9 \text{ V}$ $VM \leq VBAT - 4 \text{ V}$
14 V ≤ VBAT ≤ 45 V				

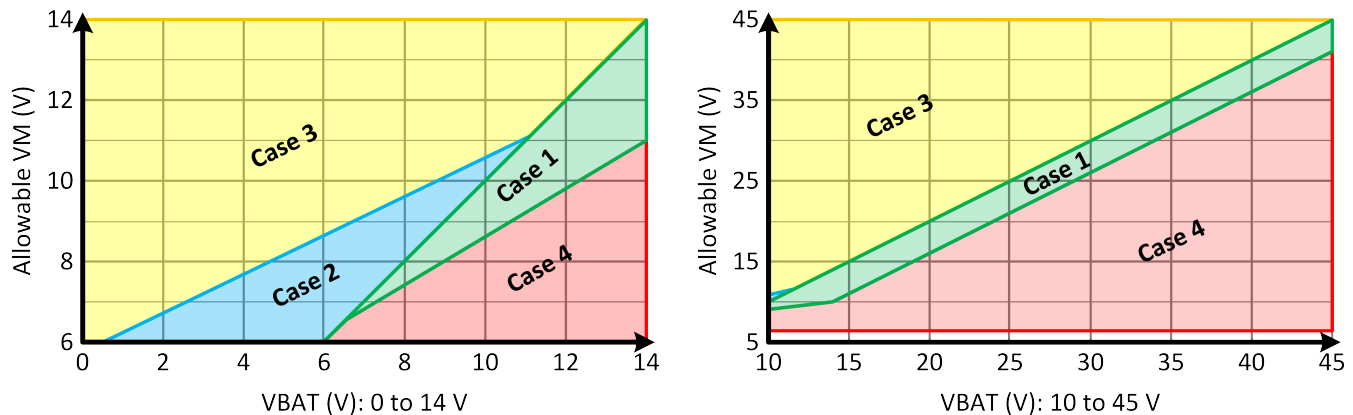


Figure 33. VM Operating Range Based on Motor Supply Voltage

When nSLEEP is low, VM may be reduced down to 0 V with up to 45 V present at VBAT. However, nSLEEP should not be brought high until VM is supplied with a voltage aligning with one of the cases outlined above.

8.2.1.2 Detailed Design Procedure

8.2.1.2.1 External FET Selection

The DRV8701 FET support is based on the charge pump capacity and output PWM frequency. For a quick calculation of FET driving capacity, use the following equations when drive and brake (slow decay) are the primary modes of operation:

$$Q_G < \frac{I_{VCP}}{f_{PWM}}$$

where

- f_{PWM} is the maximum desired PWM frequency to be applied to the DRV8701 inputs or the current chopping frequency, whichever is larger.
- I_{VCP} is the charge pump capacity, which depends on VM.

The internal current chopping frequency is at most:

$$f_{PWM} < \frac{1}{t_{OFF} + t_{BLANK}} \approx 38 \text{ kHz} \quad (6)$$

Example: If a system at VM = 7 V ($I_{VCP} = 8 \text{ mA}$) uses a maximum PWM frequency of 40 kHz, then the DRV8701 will support $Q_G < 200 \text{ nC}$ FETs.

If the application will require a forced fast decay (or alternating between drive and reverse drive), the maximum FET driving capacity is given by:

$$Q_G < \frac{I_{VCP}}{2 \times f_{PWM}} \quad (7)$$

8.2.1.2.2 IDRIIVE Configuration

Select IDRIIVE based on the gate charge of the FETs. Configure this pin so that the FET gates are charged completely during t_{DRIVE} . If the designer chooses an IDRIIVE that is too low for a given FET, then the FET may not turn on completely. TI suggests to adjust these values in-system with the required external FETs and motor to determine the best possible setting for any application.

For FETs with a known gate-to-drain charge (Q_{GD}) and desired rise time (RT), select IDRIIVE based on:

$$IDRIIVE > \frac{Q_{GD}}{RT} \quad (8)$$

Example: If the gate-to-drain charge is 2.3 nC, and the desired rise time is around 100 to 300 ns,

IDRIIVE1 = 2.3 nC / 100 ns = 23 mA

IDRIIVE2 = 2.3 nC / 300 ns = 7.7 mA

Select IDRIIVE between 7.7 and 23 mA

Select IDRIIVE as 12.5-mA source (25-mA sink)

Requires a 33-kΩ resistor from the IDRIIVE pin to GND

8.2.1.2.3 Current Chopping Configuration

The chopping current is set based on the sense resistor value and the analog voltage at VREF. Calculate the current using [Equation 9](#). The amplifier gain A_V is 20 V/V and V_{OFF} is typically 50 mV.

Example: If the desired chopping current is 3 A,

Set $R_{SENSE} = 50 \text{ m}\Omega$

$$I_{CHOP} = \frac{V_{REF} - V_{OFF}}{A_V \times R_{SENSE}} \quad (9)$$

VREF would have to be 3.05 V.

Create a resistor divider from AVDD (4.8 V) to set $V_{REF} \approx 3 \text{ V}$

Set R2 = 3.3 kΩ; set R1 = 2 kΩ.

8.2.1.3 Application Curves

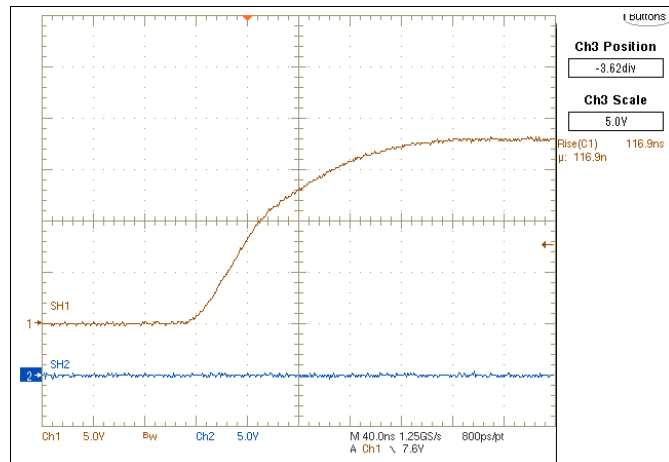


Figure 35. SH1 Rise Time (12.5-mA Source, 25-mA Sink)

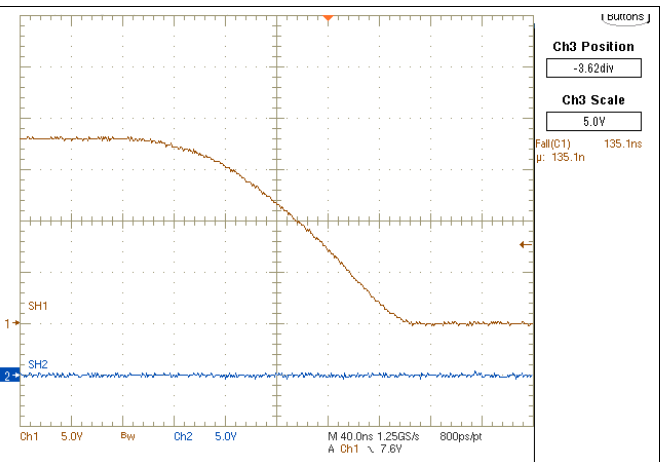


Figure 36. SH1 Fall Time (12.5-mA Source, 25-mA Sink)

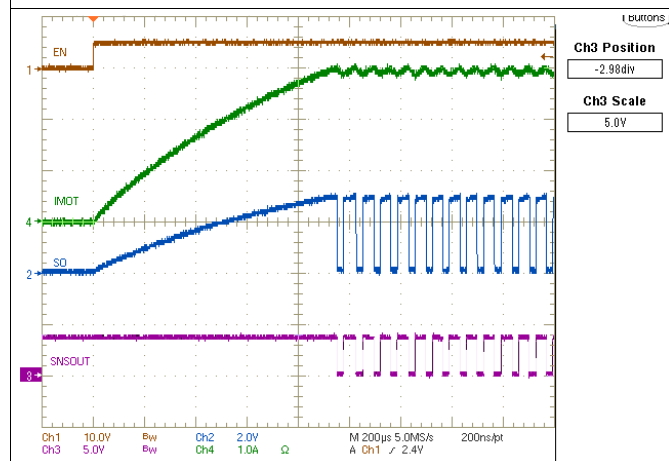


Figure 37. Current Regulating at 3 A on Motor Startup

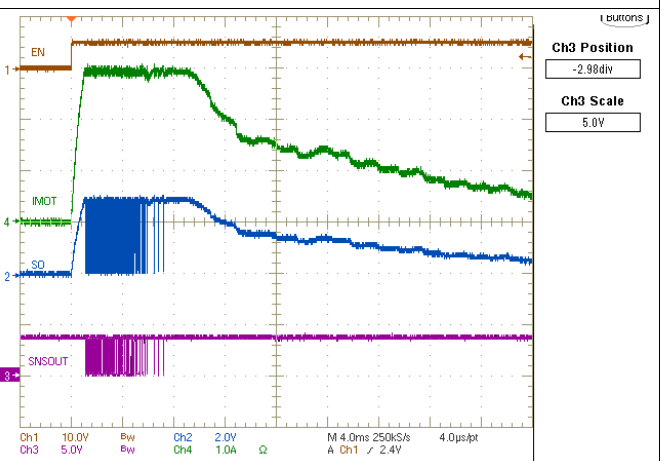


Figure 38. Current Profile on Motor Startup With Regulation

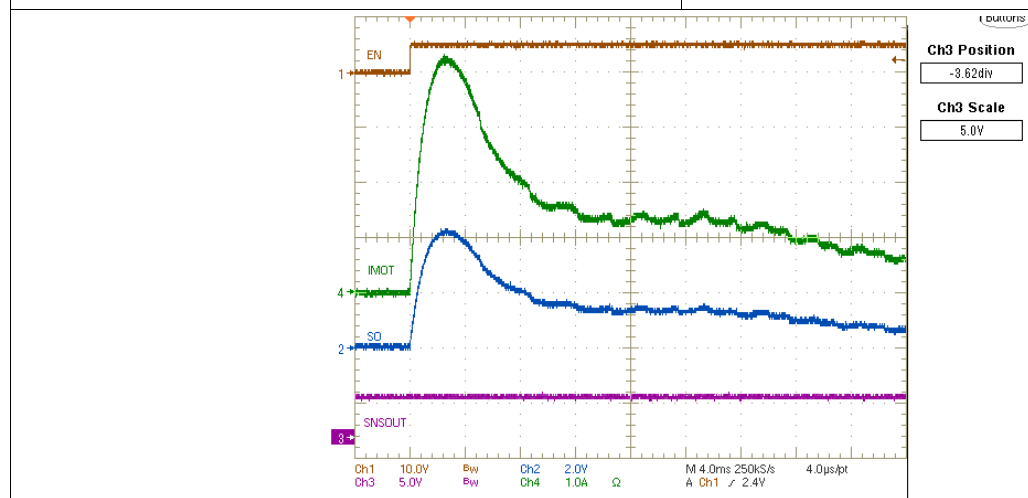


Figure 39. Current Profile on Motor Startup Without Regulation

8.2.2 Alternate Application

In this example, the DRV8701 is powered from a supply that is boosted above VBAT. This allows the system to work at lower VBAT voltages, but requires the user to disable OCP monitoring.

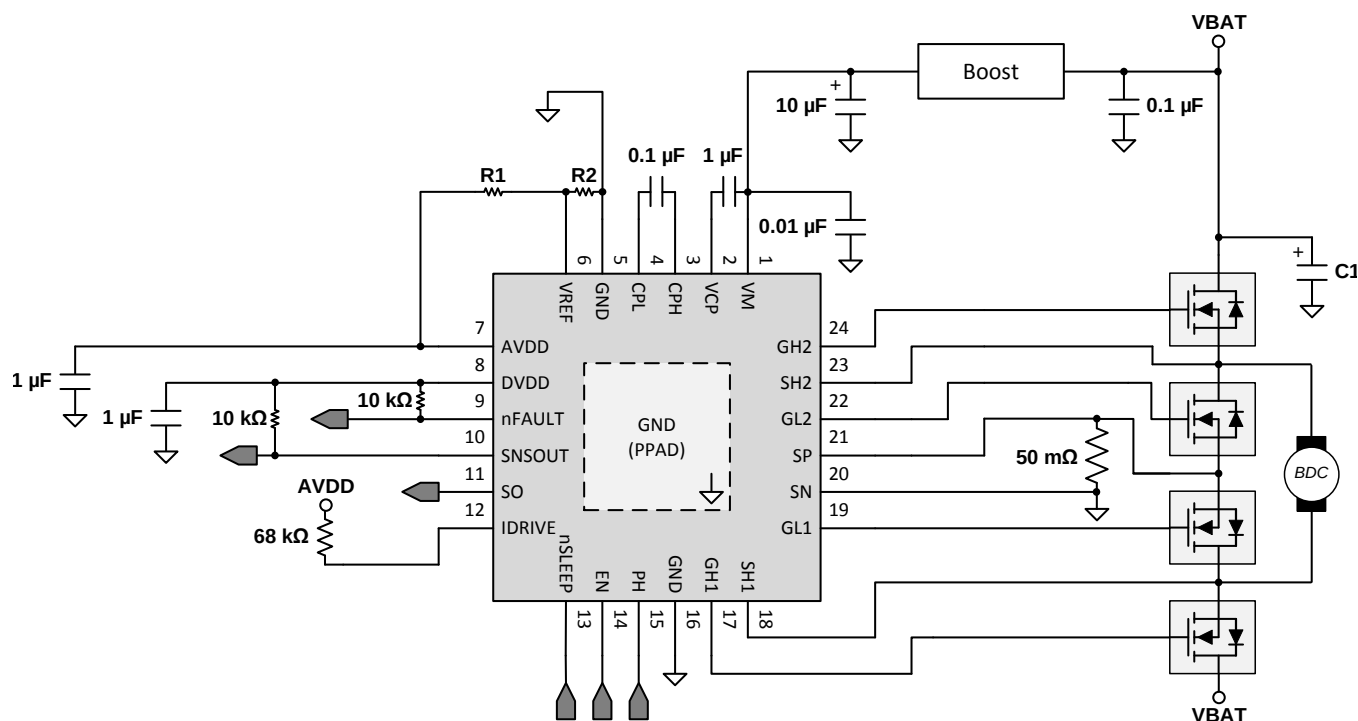


Figure 40. DRV8701 on Boosted Supply

8.2.2.1 Design Requirements

Table 11 gives design input parameters for system design.

Table 11. Design Parameters

Design Parameter	Reference	Example Value
Battery voltage	VBAT	12 V nominal Minimum operation: 4.0 V
DRV8701 supply voltage	VM	VM = 7 V when VBAT < 7 V VM = VBAT when VBAT ≥ 7 V
FET total gate charge	Q _G	42 nC
FET gate-to-drain charge	Q _{GD}	11 nC
Motor current chopping level	I _{CHOP}	3 A

8.2.3 Detailed Design Procedure

8.2.3.1 IDRIVE Configuration

Because the VM supply to the DRV8701 is different from the external H-bridge supply VBAT, the designer must disable the overcurrent monitor to prevent false overcurrent detection. The designer must place a 68-kΩ resistor between the IDRIVE pin and AVDD.

IDRIVE is fixed at 25-mA source and 50-mA sink in this mode.

So, the rise time is $11 \text{ nC} / 25 \text{ mA} = 440 \text{ ns}$.

8.2.3.2 VM Boost Voltage

To determine an effective voltage to boost VM, first determine the minimum VBAT at which the system must operate. Select VM such that the gate driver clamps do not turn on during normal operation.

$$VM < \frac{VBAT + 11.5\text{ V}}{2} \quad (10)$$

Example: If VBAT minimum is 4.0 V,

$$VM < 7.75\text{ V}$$

So VM = 7 V is selected to allow for adequate margin.

9 Power Supply Recommendations

The DRV8701 is designed to operate from an input voltage supply (VM) range between 5.9 and 45 V. A 0.1-μF ceramic capacitor rated for VM must be placed as close to the DRV8701 as possible. In addition, the designer must include a bulk capacitor with a valued of at least 10 μF on VM.

Bypassing the external H-bridge FETs requires additional bulk capacitance.

9.1 Bulk Capacitance Sizing

Having appropriate local bulk capacitance is an important factor in motor drive system design. It is generally beneficial to have more bulk capacitance, while the disadvantages are increased cost and physical size.

The amount of local capacitance needed depends on a variety of factors, including:

- The highest current required by the motor system
- The power supply's capacitance and ability to source current
- The amount of parasitic inductance between the power supply and motor system
- The acceptable voltage ripple
- The type of motor used (brushed DC, brushless DC, stepper)
- The motor braking method

The inductance between the power supply and motor drive system will limit the rate current can change from the power supply. If the local bulk capacitance is too small, the system will respond to excessive current demands or dumps from the motor with a change in voltage. When adequate bulk capacitance is used, the motor voltage remains stable and high current can be quickly supplied.

The datasheet generally provides a recommended value, but system-level testing is required to determine the appropriate sized bulk capacitor.

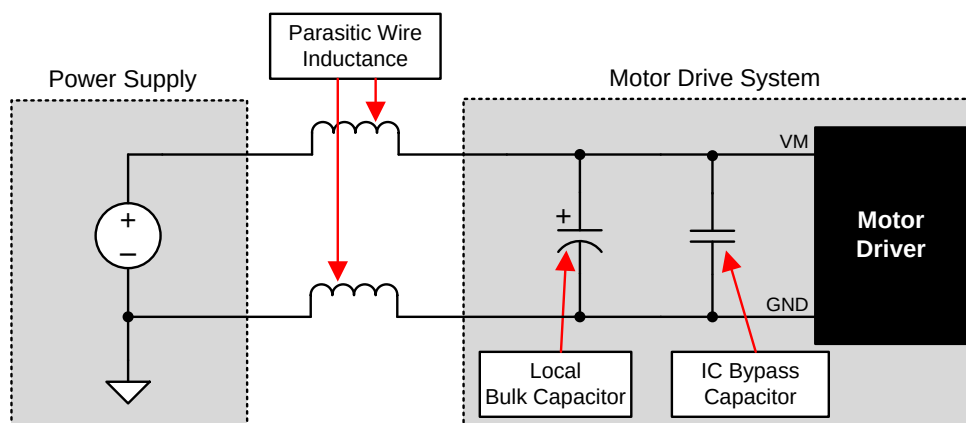


Figure 41. Example Setup of Motor Drive System With External Power Supply

The voltage rating for bulk capacitors should be higher than the operating voltage, to provide margin for cases when the motor transfers energy to the supply.

10 Layout

10.1 Layout Guidelines

Bypass the VM pin to GND using a low-ESR ceramic bypass capacitor with a recommended value of 0.1 μF rated for VM. Place this capacitor as close to the VM pin as possible with a thick trace or ground plane connection to the device GND pin.

Bypass the VM pin to ground using a bulk capacitor rated for VM. This component may be an electrolytic. This capacitance must be at least 10 μF . The bulk capacitor should be placed to minimize the distance of the high-current path through the external FETs. The connecting metal trace widths should be as wide as possible, and numerous vias should be used when connecting PCB layers. These practices minimize inductance and allow the bulk capacitor to deliver high current.

Place a low-ESR ceramic capacitor in between the CPL and CPH pins. The value for this component is 0.1 μF rated for VM. Place this component as close to the pins as possible.

Place a low-ESR ceramic capacitor in between the VM and VCP pins. The value for this component is 1 μF rated for 16 V. Place this component as close to the pins as possible.

Bypass AVDD and DVDD to ground with ceramic capacitors rated at 6.3 V. Place these bypassing capacitors as close to the pins as possible.

If desired, align the external NMOS FETs as shown in Figure 42 to facilitate layout. Route the SH2 and SH1 nets to the motor.

Use separate traces to connect the SP and SN pins to the R_{SENSE} terminals.

10.2 Layout Example

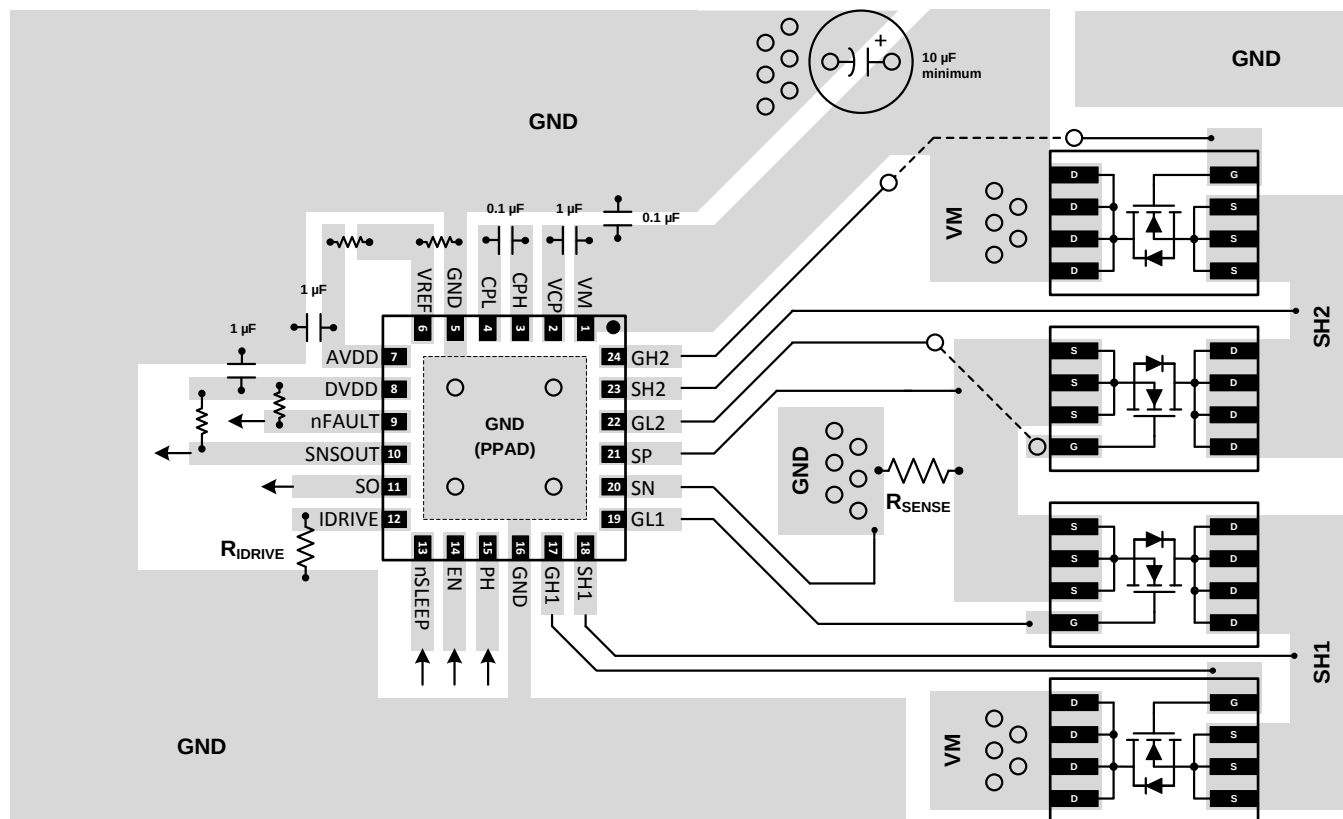


Figure 42. Layout Recommendation

11 器件和文档支持

11.1 文档支持

11.1.1 相关文档

- 《PowerPAD™ 耐热增强型封装》(, [SLMA002](#))
- 《PowerPAD™ 速成》, [SLMA004](#)
- 《电流再循环和衰减模式》, [SLVA321](#)

11.2 社区资源

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Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

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11.5 术语表

[SLYZ022](#) — *TI 术语表*。

这份术语表列出并解释术语、首字母缩略词和定义。

12 机械、封装和可订购信息

以下页中包括机械、封装和可订购信息。 这些信息是针对指定器件可提供的最新数据。 这些数据会在无通知且不
对本文档进行修订的情况下发生改变。 欲获得该数据表的浏览器版本，请查阅左侧的导航栏。

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
DRV8701ERGER	Active	Production	VQFN (RGE) 24	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	8701E
DRV8701ERGER.A	Active	Production	VQFN (RGE) 24	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	8701E
DRV8701ERGERG4	Active	Production	VQFN (RGE) 24	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	8701E
DRV8701ERGERG4.A	Active	Production	VQFN (RGE) 24	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	8701E
DRV8701ERGET	Obsolete	Production	VQFN (RGE) 24	-	-	Call TI	Call TI	-40 to 125	8701E
DRV8701PRGER	Active	Production	VQFN (RGE) 24	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	8701P
DRV8701PRGER.A	Active	Production	VQFN (RGE) 24	3000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 125	8701P
DRV8701PRGET	Obsolete	Production	VQFN (RGE) 24	-	-	Call TI	Call TI	-40 to 125	8701P

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

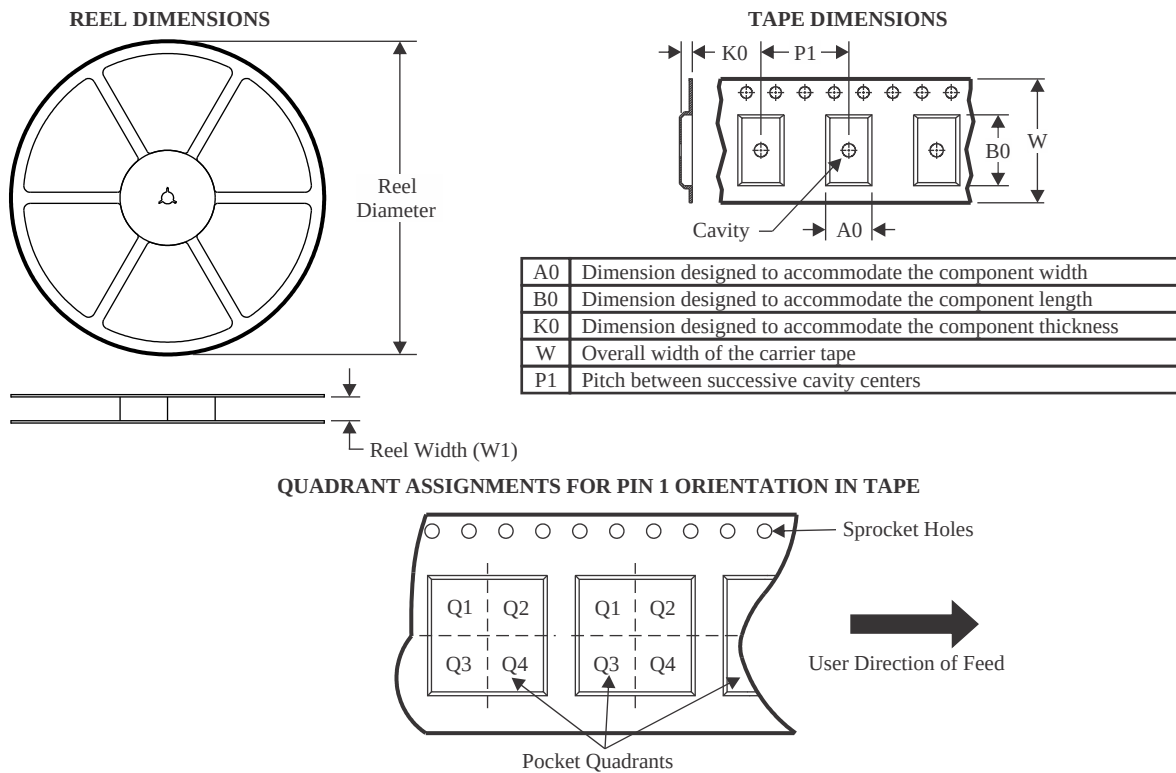
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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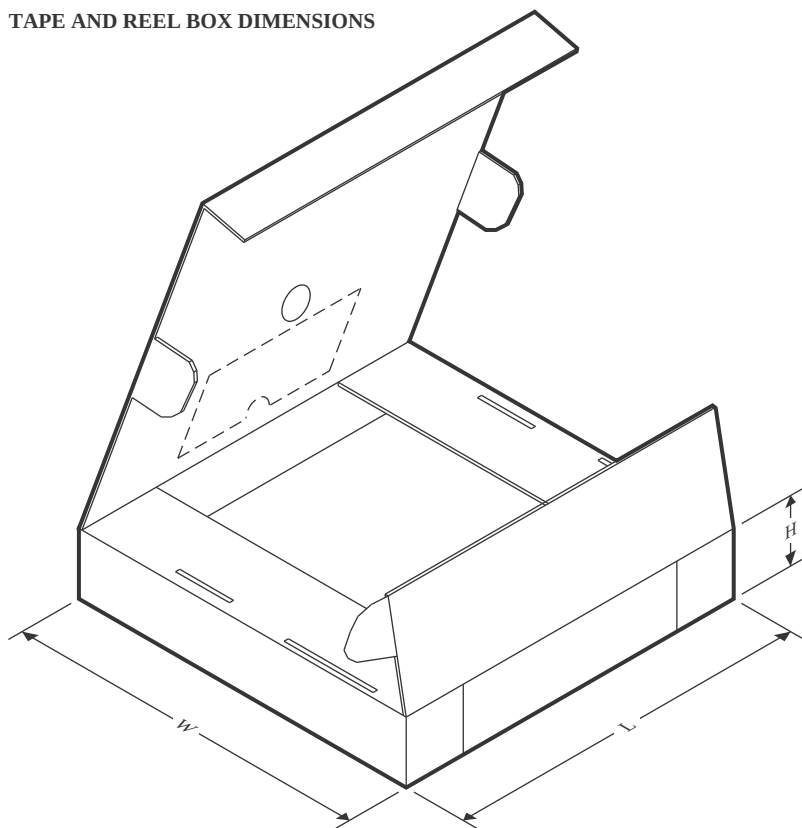
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
DRV8701ERGER	VQFN	RGE	24	3000	330.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2
DRV8701ERGERG4	VQFN	RGE	24	3000	330.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2
DRV8701PRGER	VQFN	RGE	24	3000	330.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

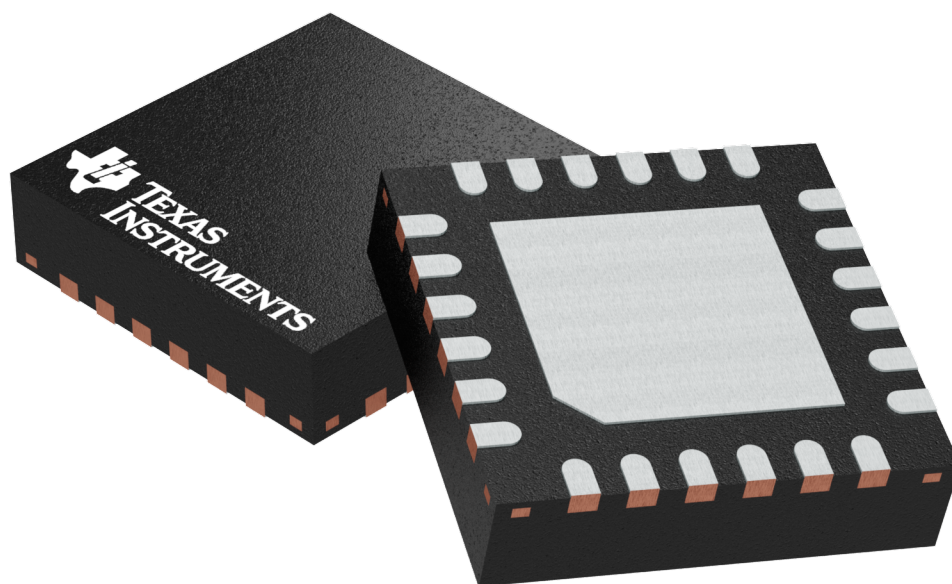
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
DRV8701ERGER	VQFN	RGE	24	3000	346.0	346.0	33.0
DRV8701ERGERG4	VQFN	RGE	24	3000	346.0	346.0	33.0
DRV8701PRGER	VQFN	RGE	24	3000	346.0	346.0	33.0

RGE 24

GENERIC PACKAGE VIEW

VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4204104/H

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